DAY 1 · MONDAY 16 NOVEMBER 2026

From NBS's own survey file to a clean analysis file

Foundations and reproducible data management



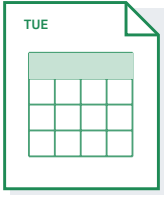
In-country Stata Training Programme
National Bureau of Statistics, Seychelles
Assoc. Prof. Endri Raço, PhD

This week you build the Q1 2026 labour market release with your own hands.

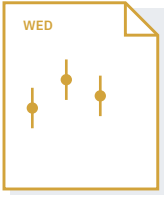
Five days, five products



Clean analysis file



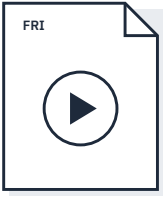
Tables pack



Official estimates



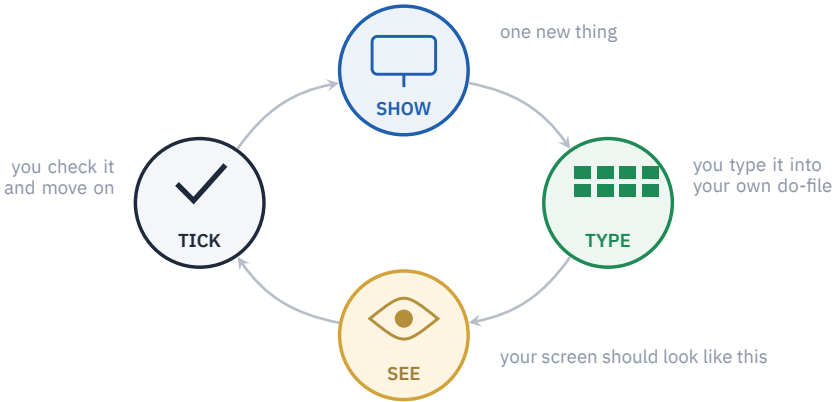
Analytical note



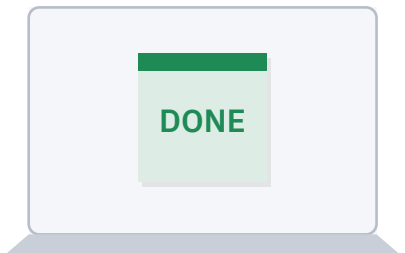
One-button release

Every step today is a step NBS takes every quarter.

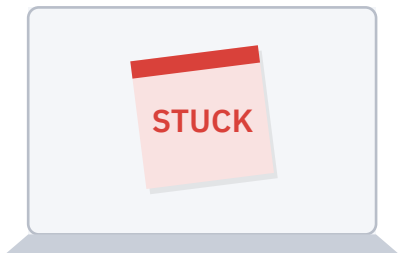
How every part of the day works



Two notes on your laptop lid



I have finished the step



please come to me

If you get lost, one file brings you back.

You will see its name at the end of each part, like this: `catchup_D1_S1.do`



1

DAY 1 · PART 1

From nothing to a first answer

One folder for the whole week



Same folders on every machine:
nothing breaks later
because of where a file lives.

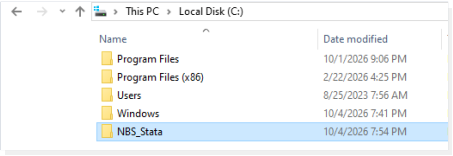
YOUR TURN · TASK 1

Create your project folder

DO

1. Open File Explorer (**Windows key + E**).
2. Click **This PC**, then **Local Disk (C:)**.
3. Right-click an empty area: **New** → **Folder**.
4. Name it `NBS_Stata` exactly.

YOU SHOULD SEE



IF NOT

Windows will not let you create a folder on C:? Put up the red note.

YOUR TURN · TASK 2

Create the eight subfolders

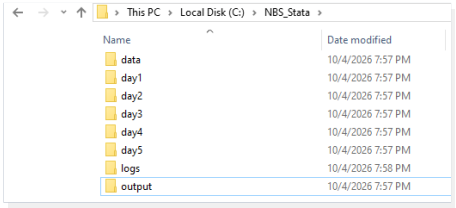
DO

Inside `NBS_Stata`, create:

`day1` `day2` `day3` `day4` `day5`
`data` `output` `logs`

Small letters, no spaces.

YOU SHOULD SEE



IF NOT

A folder is named `New folder`?
Right-click it, **Rename**.

EXTRA Sort the folders by name and check that `day1` to `day5` sit together.

YOUR TURN · TASK 3

Copy the survey files

DO

1. Open the USB stick.
2. Open its `data` folder and select all files (**Ctrl + A**).
3. Copy (**Ctrl + C**) and paste (**Ctrl + V**) into `C:\NBS_Stata\data`.

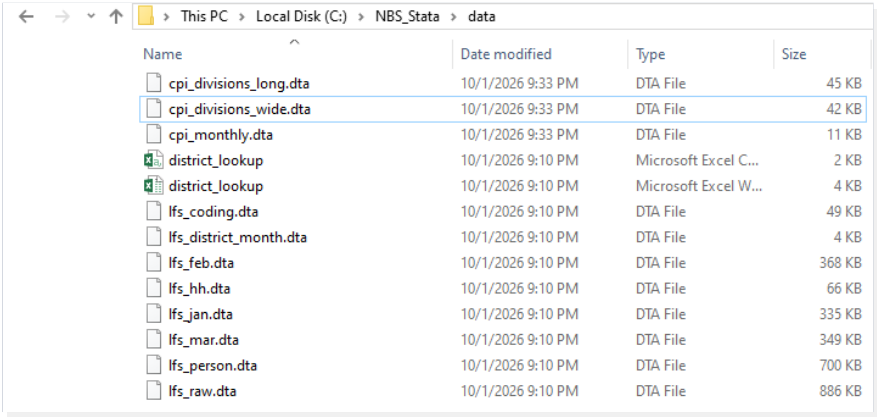
YOU SHOULD SEE

The file `lfs_raw.dta` inside `data`, with the other survey files.

IF NOT

The USB stick does not open? Put up the red note; a spare stick is on the front table.

Your folder should now look like this



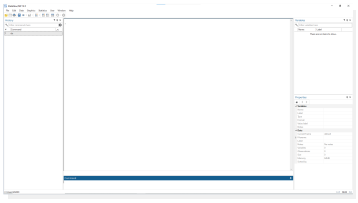
YOUR TURN · TASK 4

Open Stata

DO

1. Click the Windows **Start** button.
2. Type `Stata`.
3. Click **Stata** (the blue icon).

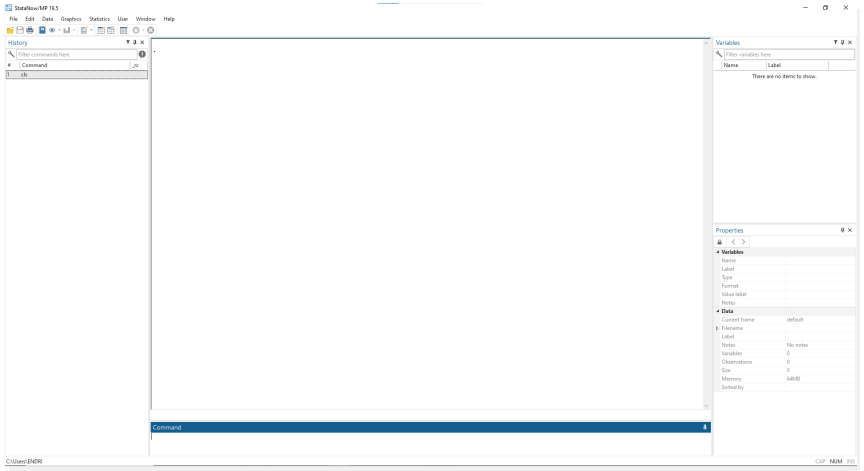
YOU SHOULD SEE



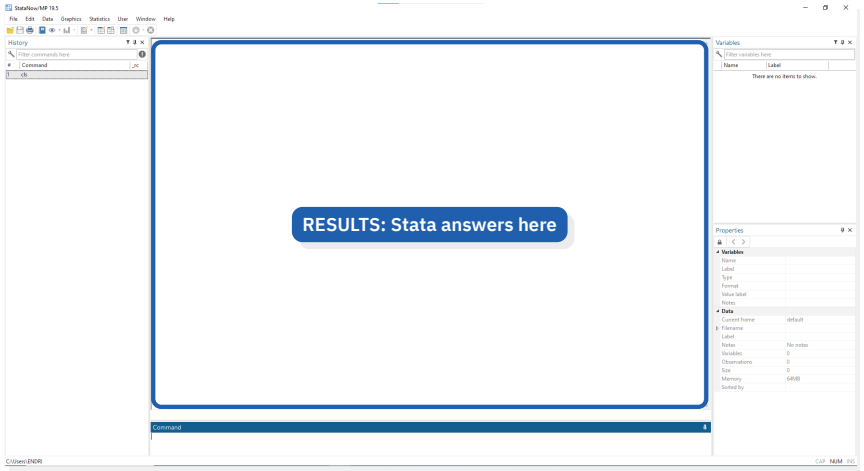
IF NOT

Several Stata icons? Take the one with the highest version number. Nothing opens after a minute: red note.

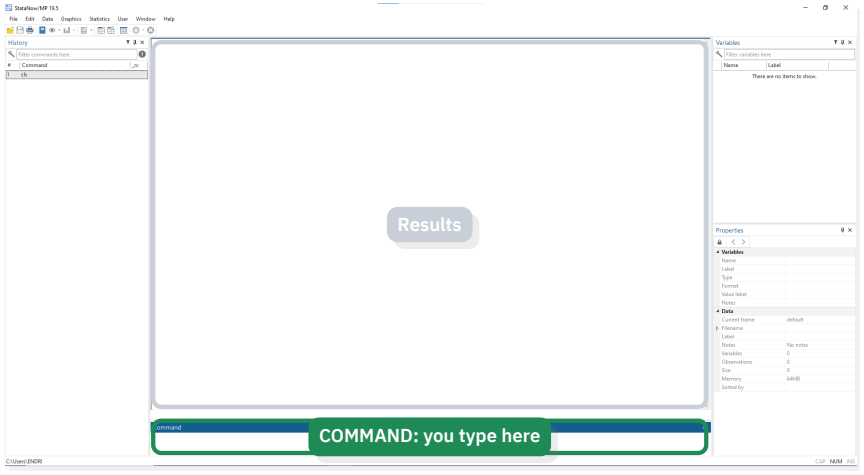
The Stata window



The Stata window



The Stata window



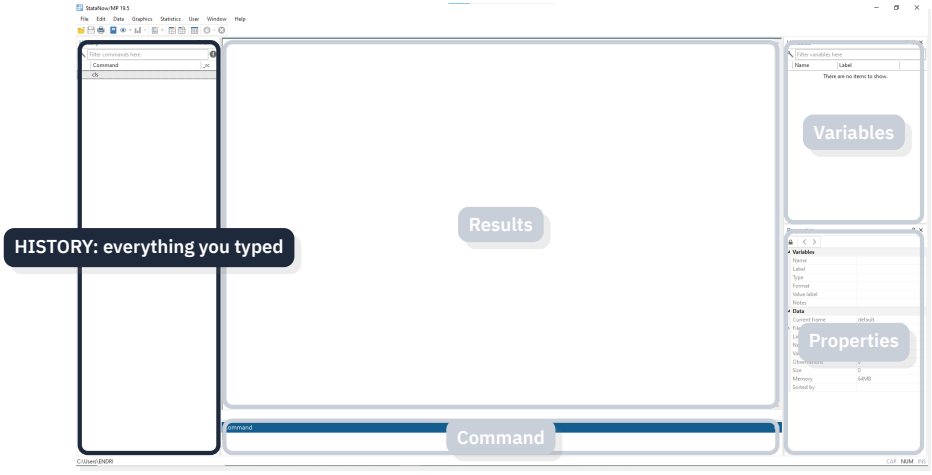
The Stata window



The Stata window



The Stata window



YOUR TURN · TASK 5

Find the five windows on your screen

DO

Point to each one, in this order:

Results → **Command** → **Variables** → **Properties** → **History**

YOU SHOULD SEE

Your screen matches the picture on the previous slides.

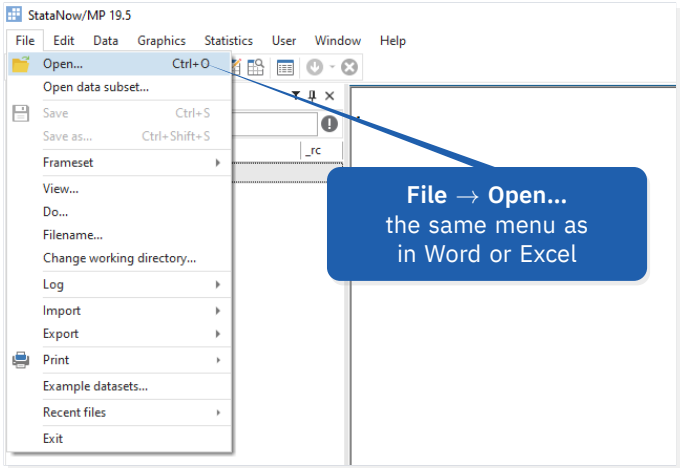
IF NOT

A window is missing? In the menu: **Window** and click its name. Still missing: red note.

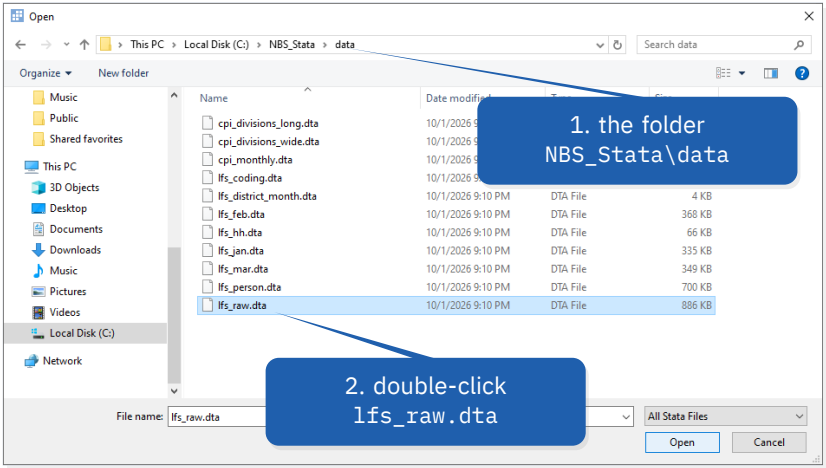
EXTRA Drag the edge between Results and Variables to make Results wider.

Now open NBS's own survey, without typing anything.

Open a file from the menu



Go to your data folder



YOUR TURN · TASK 6

Open the Q1 2026 survey file

DO

1. In Stata: **File** → **Open...**
2. Go to `C:\NBS_Stata\data`.
3. Double-click `lfs_raw.dta`.

YOU SHOULD SEE

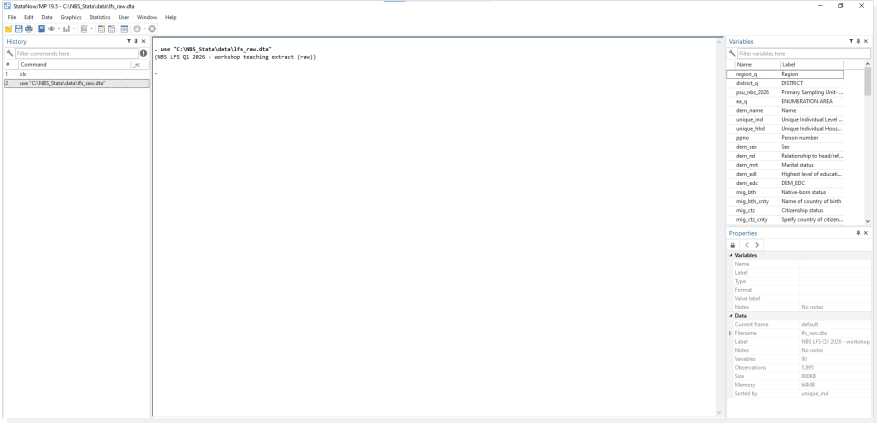
The Variables window fills with names such as `ilo_sex` and `ilo_age`.

IF NOT

`lfs_raw.dta` not in the list? Check the folder at the top of the dialog. Still missing: red note.

EXTRA Scroll the Variables window. Count how many names you recognise from the questionnaire.

Your screen should now look like this



Stata wrote a command for you

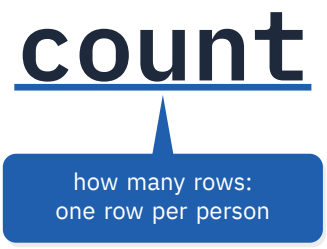
```
use "C:\NBS_Stata\data\lfs_raw.dta"
```

Stata turned your clicks into a command

Every click in a menu becomes a line like this in the Results window.

You know these variables from the questionnaire.
Here they are as columns.

Count the people in Q1 2026:



YOUR TURN · TASK 7

Count the people in the file

DO

1. Click in the Command window.
2. Type `count`
3. Press **Enter**.

YOU SHOULD SEE

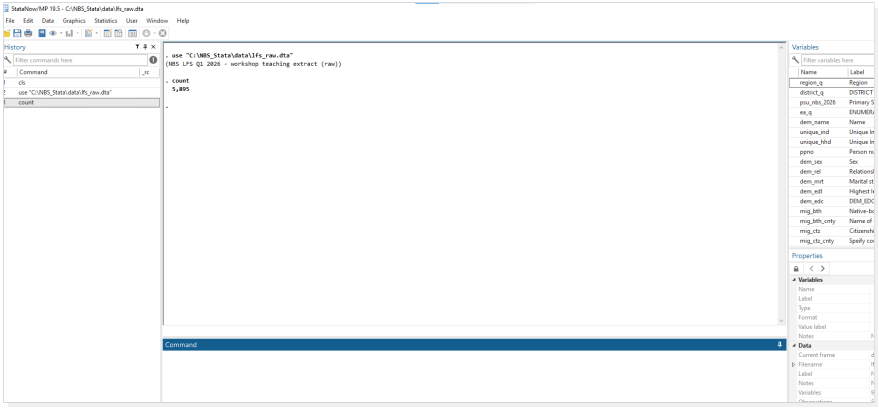
One number in the Results window, under the line `. count`

IF NOT

Red text `command not found`? Check the spelling, small letters only.

EXTRA Find the same number in the Properties window: **Data** → **Observations**.

Your result


StataNow/MP 18.5 - C:\NBS_Stata\data\lfs_raw.dta
File Edit Data Graphics Statistics User Window Help
History
Filter commands here
Command _rc
cls
use "C:\NBS_Stata\data\lfs_raw.dta"
count
. use "C:\NBS_Stata\data\lfs_raw.dta"
(NBS LFS Q1 2026 - workshop teaching extract (raw))
. count
5,895
.
Command
Variables
Filter variables here
Name Label
region_q Region
district_q DISTRICT
psu_nbs_2026 Primary S
ea_q ENUMERA
dem_name Name
unique_ind Unique in
unique_hhd Unique in
ppno Person nu
dem_sex Sex
dem_rel Relationsh
dem_mrt Marital st
dem_edl Highest le
dem_edc DBM_EDC
mig_bth Native-bo
mig_bth_cnty Name of
mig_ctz Citizenshi
mig_ctz_cnty Specify co
Properties
Variables
Name
Label
Type
Format
Value label
Notes
Data
Current frame
Filename
Label
Notes
Variables

QUICK CHECK

`count` says 5,895. What is Stata counting?

A. persons in the file: rows

B. households

C. variables (columns)

ANSWER

count says 5,895. What is Stata counting?

A persons in the file: rows

B households

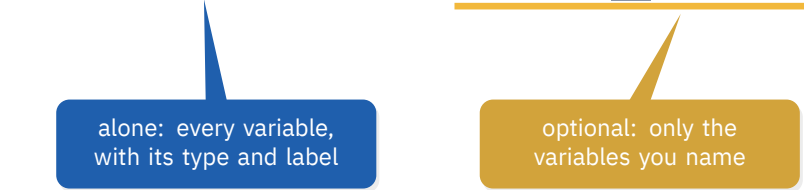
C variables (columns)

One row is one person record. Keep the number in mind: in Part 8 we find that 7 of these rows are copies, and the clean file has 5,888.

Every row is one person interviewed in Q1 2026.

See what is in the file:

`describe` `ilo_lfs19`



YOUR TURN · TASK 8

Describe the file

DO

1. Click in the Command window.
2. Type `describe`
3. Press **Enter**.

YOU SHOULD SEE

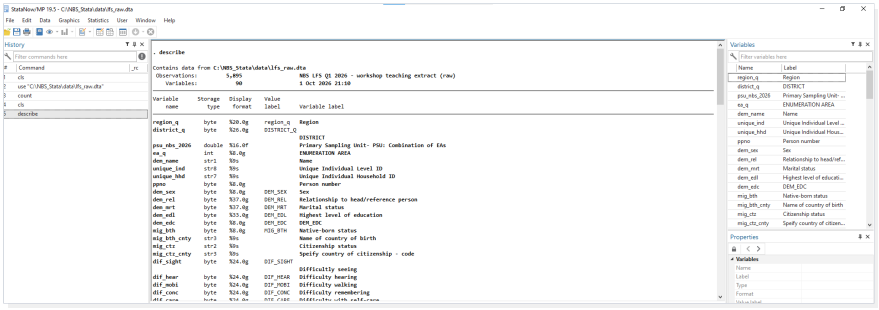
A header with **Observations** and **Variables**, then one line per variable.

IF NOT

The list ran past the top of the window? Scroll up in the Results window.

EXTRA Type `describe ilo_lfs19`. Which value label does it use?

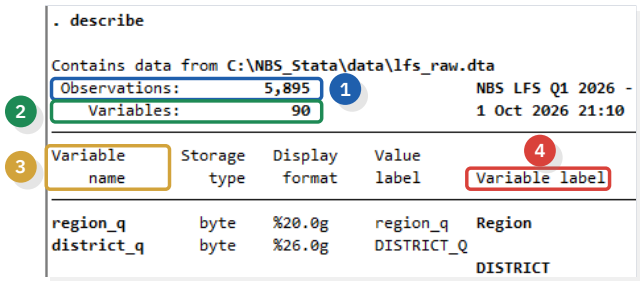
Your result


StataNow/MP 19.5 - C:\NBS_Stata\data\lfs_raw.dta
File Edit Data Graphics Statistics User Window Help
History
Filter commands here
Command
cb
use "C:\NBS_Stata\data\lfs_raw.dta"
count
cb
describe
. describe
Contains data from C:\NBS_Stata\data\lfs_raw.dta
Observations: 5,895
Variables: 90
NBS LFS Q1 2026 - workshop teaching extract (raw)
1 Oct 2026 21:10
Variable Storage Display Value
name type format label Variable label
region_q byte %20.0g region_q Region
district_q byte %26.0g DISTRICT_Q DISTRICT
psu_ebs_2026 double %16.0f Primary Sampling Unit- PSU: Combination of EAs
ea_q int %2.0g ENUMERATION AREA
dem_name str1 %9s Name
unique_ind str8 %9s Unique Individual Level ID
unique_hhd str7 %9s Unique Individual Household ID
ppno byte %8.0g Person number
dem_sex byte %8.0g DEM_SEX Sex
dem_rel byte %37.0g DEM_REL Relationship to head/reference person
dem_mrt byte %37.0g DEM_MRT Marital status
dem_edl byte %35.0g DEM_EDL Highest level of education
dem_edc byte %8.0g DEM_EDC DEM_EDC
mig_bth byte %8.0g MIG_BTH Native-born status
mig_bth_cnty str3 %9s Name of country of birth
mig_ctz str2 %9s Citizenship status
mig_ctz_cnty str3 %9s Specify country of citizenship - code
dif_sight byte %24.0g DIF_SIGHT Difficulty seeing
dif_hear byte %24.0g DIF_HEAR Difficulty hearing
dif_mobi byte %24.0g DIF_MOBI Difficulty walking
dif_conc byte %24.0g DIF_CONC Difficulty remembering
Variables
Filter variables here
Name Label
region_q Region
district_q DISTRICT
psu_ebs_2026 Primary Sampling Unit-...
ea_q ENUMERATION AREA
dem_name Name
unique_ind Unique Individual Level ...
unique_hhd Unique Individual Hous...
ppno Person number
dem_sex Sex
dem_rel Relationship to head/ref...
dem_mrt Marital status
dem_edl Highest level of educati...
dem_edc DEM_EDC
mig_bth Native-born status
mig_bth_cnty Name of country of birth
mig_ctz Citizenship status
mig_ctz_cnty Specify country of citizen...
Properties
Variables
Name
Label
Type
Format
Value label

Read the header

```
describe
```

1. **Observations**: rows, 5,895
2. **Variables**: columns, 90
3. **Variable name**: what you type
4. **Variable label**: what it means



describe, short: the file in four lines

```
. describe, short

Contains data from data/lfs_raw.dta
 Observations:         5,895                  NBS LFS Q1 2026 - workshop teaching extract (raw) 1
    Variables:            90                  1 Oct 2026 21:10 2
Sorted by: unique_ind 3
```

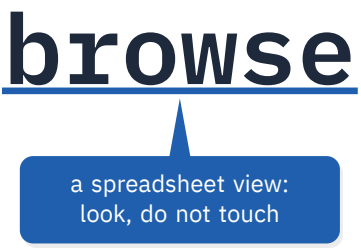
1. Observations: the rows, 5,895
2. Variables: the columns, 90
3. Sorted by: the order of the rows
4. on the right: the data label NBS gave the file

Read three variables

```
Variable      Storage   Display    Value
    name         type    format    label      Variable label
-------------------------------------------------------------------------------
ilo_lfs19       byte    %24.0g     label_ilo_lfs19
                                              Labour Force Status - 19th ICLS
dem_sex         byte    %8.0g      DEM_SEX    Sex
ilo_age         byte    %9.0g                 Age in completed years (top-coded at 85)
```

1. Storage type: `byte` is a small whole number
2. a variable label: what the variable means
3. Value label: the dictionary of codes (`DEM_SEX`)
4. `ilo_age` has no value label: ages are numbers, not codes

Look at the data, row by row:



YOUR TURN · TASK 9

Browse the file

DO

1. Type `browse` and press **Enter**.
2. Look around: rows and columns.
3. Close the window when done.

YOU SHOULD SEE

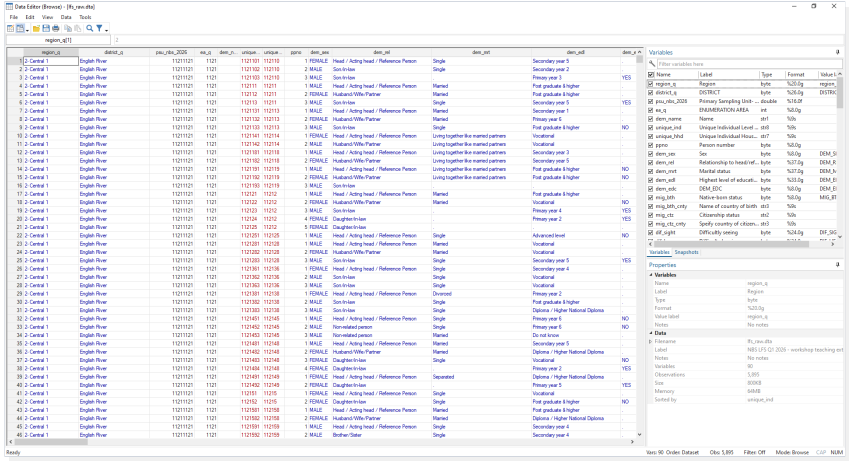
A new window, the Data Editor in **Browse** mode, with the file as a table.

IF NOT

The title says **Edit** mode? Close it and type `browse` again.

EXTRA Find the column `ilo_sex`. Click one cell: which number is behind the word?

Your result


Data Editor (Browse) - [ifs_raw.dta]
File Edit View Data Tools
region_s[1]
Variables
Filter variables here
Name Label Type Format Value L
region_s Region byte %26.0g region_
district_q DISTRICT byte %26.0g DISTRIC
psu_nbs_2026 Primary Sampling Unit - ... double %16.0f
ea_q ENUMERATION AREA int %8.0g
dem_name Name str1 %9s
unique_ind Unique Individual Level ... str9 %9s
unique_hhd Unique Individual Hous... str7 %9s
ppno Person number byte %8.0g
dem_sex Sex byte %8.0g DEM_SI
dem_rel Relationship to head/ref... byte %37.0g DEM_R
dem_mrt Marital status byte %37.0g DEM_M
dem_edl Highest level of educati... byte %33.0g DEM_E
dem_edc DEM_EDC byte %8.0g DEM_E
eng_bth Native-born status byte %8.0g MIG_BT
eng_bth_cnty Name of country of birth str3 %9s
eng_cts Citizenship status str2 %9s
eng_cts_cnty Spoty country of citizen... str3 %9s
dif_sight Difficultly seeing byte %24.0g DIF_SIG
Variables Snapshots
Properties
Variables
Name region_q
Label Region
Type byte
Format %20.0g
Value label region_q
Notes No notes
Data
Filename ifs_raw.dta
Label NBS | FS Q1 2026 - workshop teaching ext
Notes No notes
Variables 90
Observations 5,895
Size 800KB
Memory 64MB
Sorted by unique_ind
Ready
Vars: 90 Order: Dataset Obs: 5,895 Filter: Off Mode: Browse CAP NUM

Three colours in the Data Editor

	region_q	district_q	psu_nbs_2026	ea_q	dem_n...	unique...	unique...
1	2- Central 1	English River	11211121	1121		1121101	112110
2	2- Central 1	English River	11211121	1121		1121102	112110
3	2- Central 1	English River	11211121	1121		1121103	112110
4	2- Central 1	English River	11211121	1121		112111	11211
5	2- Central 1	English River	11211121	1121		112112	11211
6	2- Central 1	English River	11211121	1121		112113	11211

1121 a number **English River** a label **1121101** text

One row is one person. Blue text has a number underneath: click a blue cell to see it.

RULE

Browse, never edit.

A value changed by hand leaves no trace,
and nobody can repeat it next quarter.

Every change is a line in a do-file.

Count the people in each category:



YOUR TURN · TASK 10

Count men and women

DO

1. Type `tabulate ilo_sex`
2. Press **Enter**.

YOU SHOULD SEE

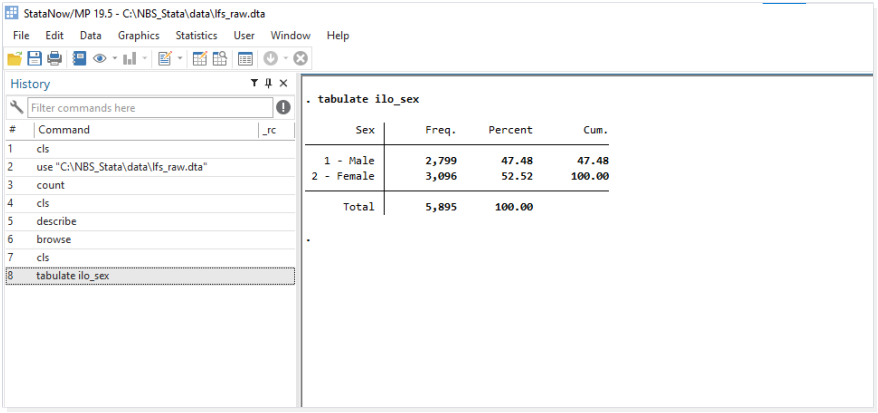
A small table: **Male** and **Female**, with Freq., Percent and Cum., and a Total row.

IF NOT

Red text `variable not found`? Check the name: `ilo`, underscore, `sex`.

EXTRA Type `tabulate region_q`. Which region has the most people?

Your result


StataNow/MP 19.5 - C:\NBS_Stata\data\lfs_raw.dta
File Edit Data Graphics Statistics User Window Help
History
Filter commands here
Command _rc
1 cls
2 use "C:\NBS_Stata\data\lfs_raw.dta"
3 count
4 cls
5 describe
6 browse
7 cls
8 tabulate ilo_sex
. tabulate ilo_sex
Sex Freq. Percent Cum.
1 - Male 2,799 47.48 47.48
2 - Female 3,096 52.52 100.00
Total 5,895 100.00
.

ON YOUR OWN · A

Praslin and La Digue

THE QUESTION

How many records come from region 6, Praslin & La Digue? What share of the file is that?

HINT

One variable, one command, the one you just used. The region variable is `region_q`.

ON YOUR OWN · A · ANSWER

```
tabulate region_q
```

```
              Region |      Freq.     Percent        Cum.
---------------------+-----------------------------------
           1- North |         943       16.00       16.00
         2- Central 1 |         882       14.96       30.96
         3- Central 2 |         843       14.30       45.26
         4- EastSouth |       1,451       24.61       69.87
             5- West |         900       15.27       85.14
6- Praslin & LaDigue |         514        8.72       93.86  (1)
     7- Perseverance |         362        6.14      100.00
---------------------+-----------------------------------
               Total |       5,895      100.00
```

(1) 514 records, 8.72% of the file

COMMON ERROR

A capital letter in a command

```
. Tabulate ilo_sex
command Tabulate is unrecognized
```

WHY

Stata reads commands exactly as typed. `Tabulate` and `tabulate` are different words, and only one exists.

FIX

Commands and variable names are lower case: `tabulate ilo_sex`.

You can now

- ✓ **`use`** open the Q1 2026 survey file
- ✓ **`count`** count its records
- ✓ **`describe`** list what is in it
- ✓ **`tabulate`** count people in each category

Four commands, four facts about Seychelles' labour force survey.

Everything you typed is lost if Stata closes.

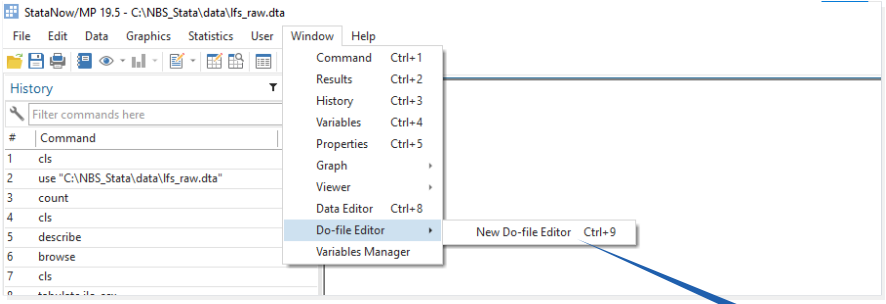
Two files make work repeatable



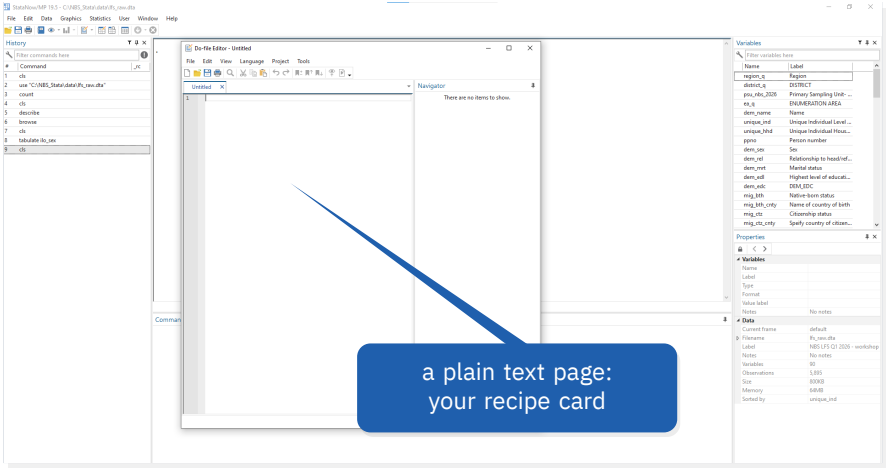
do-file
what to do, step by step

log
what happened, with results

Open the Do-file Editor



The Do-file Editor



YOUR TURN · TASK 11

Open the Do-file Editor

DO

1. Press **Ctrl + 9**.
2. Place the editor beside the main window.

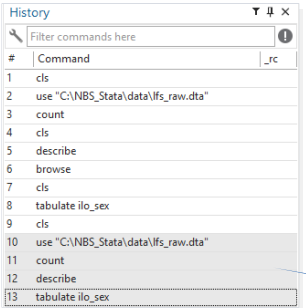
YOU SHOULD SEE

A new window, **Do-file Editor - Untitled**, with an empty page and line number 1.

IF NOT

Nothing opens? Use the menu: **Window** → **Do-file Editor** → **New Do-file Editor**.

Send your commands to the editor



select the four lines, right-click, **Send to Do-file Editor**

YOUR TURN · TASK 12

Send your commands to the editor

DO

1. In **History**, hold **Ctrl** and click the `use`, `count`, `describe` and `tabulate` lines.
2. Right-click one of them.
3. Click **Send to Do-file Editor**.

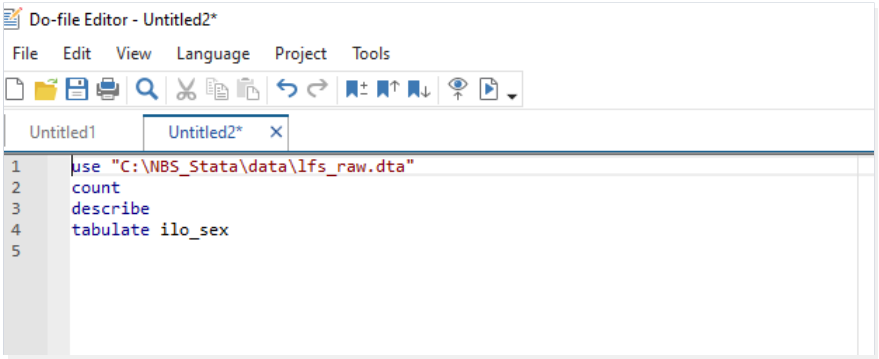
YOU SHOULD SEE

The four lines in the Do-file Editor, one per line.

IF NOT

A `cls` or `browse` line came along? Delete that line in the editor.

Your editor should now look like this



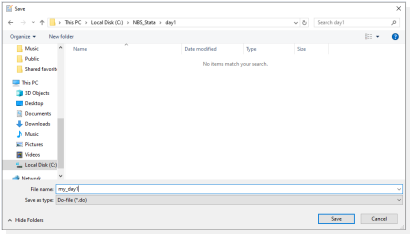
YOUR TURN · TASK 13

Save it in your day1 folder

DO

1. In the editor: **File** → **Save As...**
2. Go to `C:\NBS_Stata\day1`.
3. File name: `my_day1.do`, then **Save**.

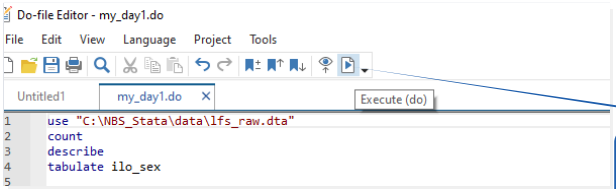
YOU SHOULD SEE



IF NOT

The tab still says **Untitled**? Save again and check the folder.

Run the file



Execute (do)
the same as Ctrl + D

`Ctrl + D` runs the **whole file**

`Ctrl + D` on **selected lines**: runs only those

YOUR TURN · TASK 14

Run your do-file

DO

1. Click inside the editor.
2. Press **Ctrl + D**.

YOU SHOULD SEE

The four commands run again in Results, in order, and the last line says `end of do-file`.

IF NOT

Nothing happens? Click inside the editor first, then Ctrl + D.

EXTRA Select only the `tabulate` line and press Ctrl + D.

Your result

```
Sorted by: unique_ind

. tabulate ilo_sex

        Sex |      Freq.     Percent        Cum.
------------+-----------------------------------
     1 - Male |      2,799       47.48       47.48
  2 - Female |      3,096       52.52      100.00
------------+-----------------------------------
      Total |      5,895      100.00

.
end of do-file

.
```

In one hour you opened NBS's own survey, answered four questions and wrote your first reproducible program.

lost? open **catchup_D1_S1.do**

Fifteen short questions.
They tell me where to spend time this week.
It is not a test of you.

Entry diagnostic

1. Open the link or scan the code.
2. Enter your **participant code**, not your name.
3. Answer all 15 questions, then **Submit**.

No Wi-Fi? Take a paper copy from the front table.



forms.gle/FizcUZxFoLxqsaL59

Done? Put up your green note.



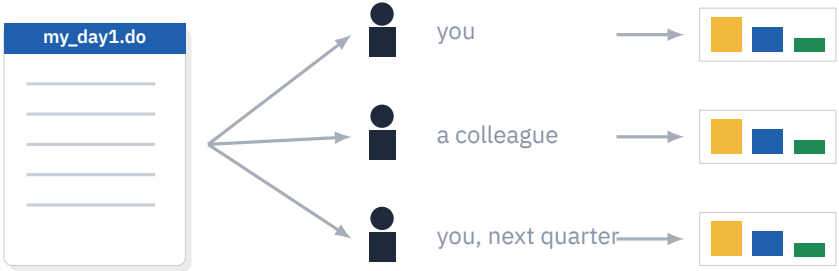
2

DAY 1 · PART 2

Work that can be repeated

A result NBS publishes must be reproducible by a colleague, next quarter, from the files alone.

One file, the same result for everyone



Same file, same data: same table.

Write notes for your colleague:

```
* my_day1.do: Day 1, NBS LFS Q1 2026
count  // number of persons in the file
```

`*` at the start:
the whole line is a note

`//` after a command:
a note on that line

Stata skips comments. People read them.

YOUR TURN · TASK 15

Sign your do-file

DO

1. Click at the start of line 1 and press **Enter** to make room.
2. On the new line 1, type a comment: `* my_day1.do, your name, 16 Nov 2026`
3. Save with **Ctrl + S**.

EXTRA Add a `//` note after the `count` line.

YOU SHOULD SEE

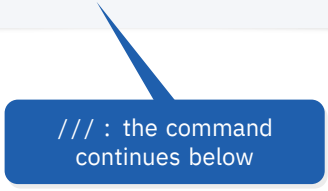
The new line turns green.

IF NOT

The line stays black? Check that it starts with `*`.

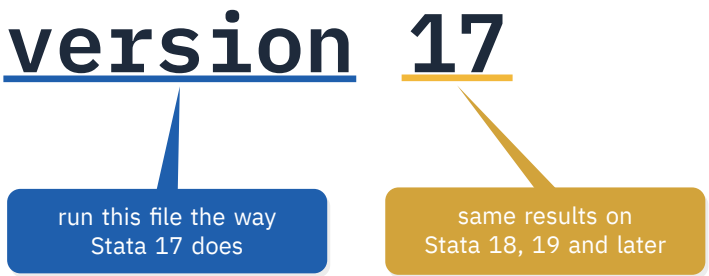
Split a long command over two lines:

```
describe ilo_sex ilo_age ///
     region_q district_q
```



Only in do-files. In the Command window, type it on one line.

Fix the Stata release at the top:



Start from an empty Stata:

```
clear all
```

empties memory:
data, labels, results

Never pause the output:

```
set more off
```

no --more-- stops in long output

Already the setting since Stata 16. You will meet it in older NBS do-files.

Tell Stata where your project lives:



RULE

Paths: forward slashes, inside quotes.

not `C:\NBS_Stata\data`

but `"C:/NBS_Stata/data"`

Forward slashes work on every machine. Quotes protect paths with spaces.

YOUR TURN · TASK 16

Add the settings at the top

DO

Under your comment line, add:

```
version 17
clear all
cd "C:/NBS_Stata"
```

Save, then run the whole file with **Ctrl + D**.

YOU SHOULD SEE

After the `cd` line, Results shows `C:\NBS_Stata`.

IF NOT

Red `unable to change to`? Check the folder name and the quotes.

Your result

```
. do "C:\NBS_Stata\day1\my_day1.do"

. * my_day1.do, Endri, 16 Nov 2026
. version 17

. clear all

. cd "C:/NBS_Stata"
C:\NBS_Stata
```

COMMON ERROR

A folder that does not exist

```
. cd "C:/NBS_Stat"
unable to change to C:/NBS_Stat
```

WHY

The path is one letter short: `NBS_Stat`. Stata stays where it was.

FIX

Copy the path from the address bar of File Explorer, then change \ to /.

Ask where you are:



Open the data the short way:

```
use "data/lfs_raw.dta", clear
```



The path starts inside `C:/NBS_Stata`, because of `cd`.

YOUR TURN · TASK 17

Use the short path

DO

1. Replace the `use` line Stata wrote with: `use "data/lfs_raw.dta", clear`
2. Save, then run the whole file with **Ctrl + D**.

YOU SHOULD SEE

The same count, describe and table as before, ending with `end of do-file`.

IF NOT

Red `file not found`? Check that the `cd` line ran first.

A clean run, top to bottom

```
ilo_job1_h~sual int     %9.0g                 Weekly hours usually worked - main job
ilo_job1_h~tual float   %9.0g                 Weekly hours actually worked - main job
ilo_joball_ho~l int     %9.0g                 Weekly hours usually worked - all jobs
ilo_job1_ife_~d byte    %33.0g      ilo_ife_prod_lab
                                              Informal / Formal Economy (Unit of production)
ilo_job1_ife_~e byte    %34.0g      ife_nature_lab
                                              Informal / Formal Economy (Nature of job) - Main job
ilo_joball_tru  byte    %26.0g      tru_lab   Time-related underemployed
ilo_dur_aggre~e byte    %35.0g      unemp_aggr
                                              Duration of unemployment (Aggregate)
ilo_dis         byte    %23.0g      ilo_dis_lab
                                              Discouraged job-seekers
ilo_neet        byte    %46.0g      neet_lab  Youth not in education, employment or training
ilo_sub_all     byte    %17.0g      sub_all_lab
                                              Own-use producers
int_month       byte    %8.0g       int_month
                                              Interview month, Q1 2026
------------------------------------------------------------------------------
Sorted by: unique_ind

.    tabulate ilo_sex

        Sex |      Freq.     Percent        Cum.
------------+-----------------------------------
   1 - Male |      2,799       47.48       47.48
 2 - Female |      3,096       52.52      100.00
------------+-----------------------------------
      Total |      5,895      100.00

.
end of do-file

.
```

COMMON ERROR

Opening a file over unsaved changes

```
. use "data/lfs_raw.dta"
no; dataset in memory has changed since last saved
```

WHY

Memory holds a changed dataset. Stata refuses to throw your changes away without being told.

FIX

Add `, clear` to `use`. In a do-file this is always what you want.

Keep a diary of every run:



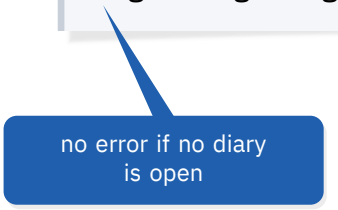
Close the diary at the end:

```
log close
```

the last line
of every do-file

Close any diary left open, first:

```
capture log close
log using "logs/day1.log", replace text
```



A run that stopped halfway leaves its diary open. This line handles that.

YOUR TURN · TASK 18

Add the diary to your do-file

DO

1. Under `cd`, add:
   ```
   capture log close
   log using "logs/day1.log", replace text
   ```
2. As the last line, add `log close`.
3. Save, then **Ctrl + D**.

EXTRA Open `logs` in File Explorer: is `day1.log` there?

YOU SHOULD SEE

Results shows a short block with `log:` and the path of `day1.log`, at the start and at the end.

IF NOT

Red `file ... could not be opened`? Check that the `logs` folder exists.

COMMON ERROR

A second log while one is open

```
. log using "_trainer/capture/second.log", text
log file already open
```

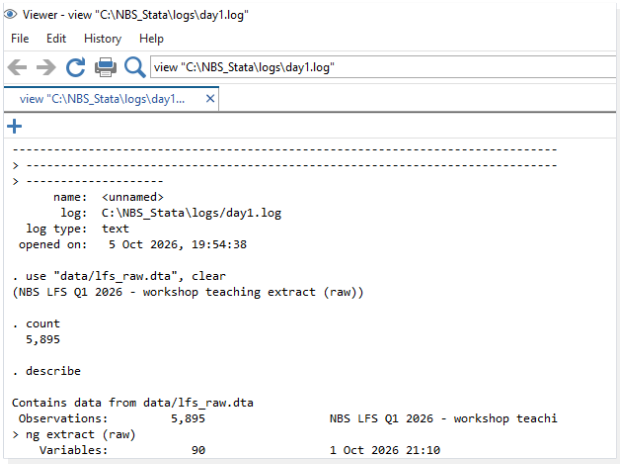
WHY

The last run stopped before `log close`. The log is still open, so the next `log using` stops.

FIX

Put `capture log close` just before `log using`, as on the last slide.

Your diary, opened in Stata



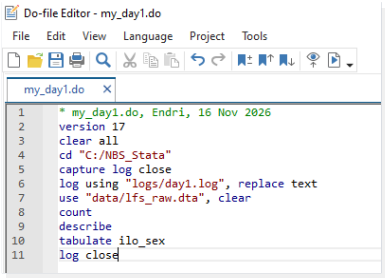
Every do-file has the same six parts

header	`* purpose, author, date`
settings	`version 17  clear all  cd`
diary opens	`capture log close  log using`
data in	`use "data/lfs_raw.dta", clear`
the work	`count  describe  tabulate`
diary closes	`log close`

Data out (`save`) joins later today, before the diary closes.

Your do-file so far

```
* my_day1.do, your name, 16 Nov 2026
version 17
clear all
cd "C:/NBS_Stata"
capture log close
log using "logs/day1.log", replace text
use "data/lfs_raw.dta", clear
count
describe
tabulate ilo_sex
log close
```



QUICK CHECK

You run my_day1.do a second time and it stops at `log using`. The most likely reason?

A. the log file already exists

B. a log is still open from the run that stopped

C. the version line is missing

ANSWER

You run my_day1.do a second time and it stops at `log using`. The most likely reason?

A. the log file already exists

B. a log is still open from the run that stopped

C. the version line is missing

An existing file is handled by `replace`. An open log is handled by `capture log close` on the line before.

Open Stata's manual, inside Stata:



YOUR TURN · TASK 19

Read the help for tabulate

DO

1. In the Command window type `help tabulate oneway`
2. Press **Enter**.
3. Scroll down to **Examples**.

YOU SHOULD SEE

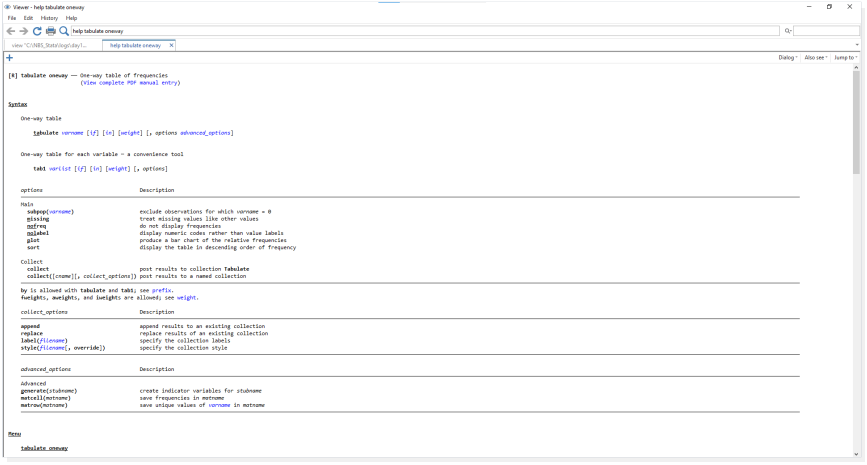
A Viewer window with the help page: Syntax, Options, Examples.

IF NOT

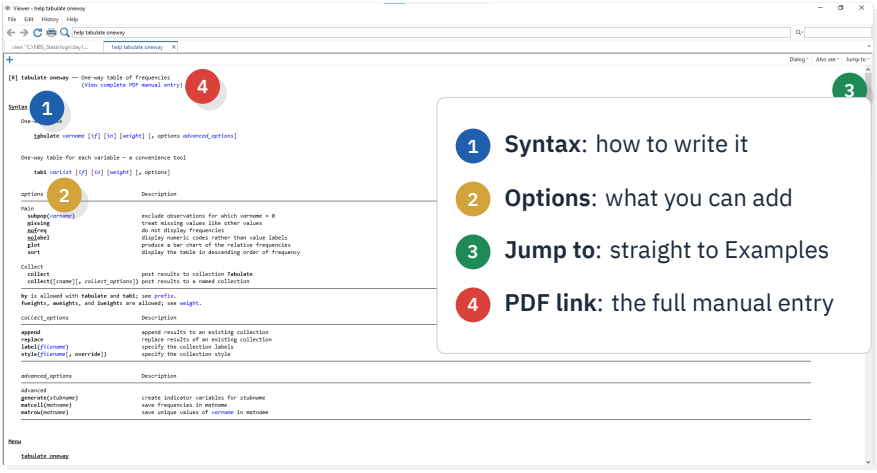
`help for tabulate not found`? Check the spelling and the space.

EXTRA Copy one example line into your do-file and run it.

Your result


Viewer - help tabulate oneway
File Edit History Help
help tabulate oneway
help tabulate oneway
Dialog Also see Jump to
[R] tabulate oneway — One-way table of frequencies
(View complete PDF manual entry)
Syntax
One-way table
tabulate varname [if] [in] [weight] [, options advanced_options]
One-way table for each variable + a convenience tool
tabi varlist [if] [in] [weight] [, options]
options Description
Main
subpop(varname) exclude observations for which varname = 0
missing treat missing values like other values
nofreq do not display frequencies
nolabel display numeric codes rather than value labels
plot produce a bar chart of the relative frequencies
sort display the table in descending order of frequency
Collect
collect post results to collection Tabulate
collect[(cname)][, collect_options]) post results to a named collection
by is allowed with tabulate and tabi; see prefix.
fweights, aweights, and iweights are allowed; see weight.
collect_options Description
append append results to an existing collection
replace replace results of an existing collection
label(filename) specify the collection labels
style(filename[, override]) specify the collection style
advanced_options Description
Advanced
generate(stubname) create indicator variables for stubname
matcell(matname) save frequencies in matname
matrow(matname) save unique values of varname in matname
Menu
tabulate oneway

Four parts of every help page



The underlined letters

[R] tabulate oneway — One-way table of frequencies
(View complete PDF manual entry)

Syntax

One-way table

tabulate *varname* [*if*] [*in*] [*weight*] [, *options advanced_options*]

One-way table for each variable — a convenience tool

tab1 *varlist* [*if*] [*in*] [*weight*] [, *options*]

`ta` is the shortest name for `tabulate`

In do-files, write the full name: a colleague reads it faster.

RULE

Red text means Stata stopped.

Read the **first line** of the red message.
It names the problem. Fix it, run again.

The code in brackets, like `r(111)`, is a link to more help.

YOUR TURN · TASK 20

Make Stata stop, then fix it

DO

1. Type `tabulate ilosex` (no underscore) and press **Enter**.
2. Read the red message.
3. Type the correct command.

YOU SHOULD SEE

Red: `variable ilosex not found.` Then the table for `ilo_sex`.

IF NOT

No red? Check that you left out the underscore.

EXTRA Click `r(111)` in the red message.

Your result

```
. tabulate ilosex
variable ilosex not found
r(111);

. tabulate ilo_sex
```

Sex	Freq.	Percent	Cum.
1 - Male	2,799	47.48	47.48
2 - Female	3,096	52.52	100.00
Total	5,895	100.00	

Parts 1 and 2 on one page

CHEAT CARD · KEEP THIS ONE

`use "data/file.dta", clear`	open a Stata file
`count`	how many rows
`describe / describe, short`	the variables / the file in four lines
`browse`	look at the data, never edit
`tabulate v`	counts of one variable
`* note / // note`	comments for the reader
`///`	continue a command on the next line
`version 17 / clear all / cd "C:/NBS_Stata"`	the top of every do-file
`capture log close`	close a log left open
`log using "logs/day1.log", replace text`	start the diary
`log close`	end the diary
`help command`	the manual page

Your work now runs from one file, leaves a diary, and works on any colleague's machine.

lost? open **catchup_D1_S2.do**

Lab 1 · First look at the Q1 file

THE SITUATION

Your section head asks: how many records, how many variables, how many men and women? Answer from a do-file, with a log.

OPEN

Copy `lab_D1_1.do` from the USB `day1` folder into `C:\NBS_Stata\day1`, then open it in the Do-file Editor.

SIX TASKS

CORE	Tasks 1 to 4
EXTENSION	Task 5
CHALLENGE	Task 6

CORRECT WHEN

The whole file runs with no red text and `logs/lab_D1_1.log` exists.

How the lab works

```
* TASK 3 (Core) How many records?
*   Count the records in the file.
* >>> YOUR CODE
count

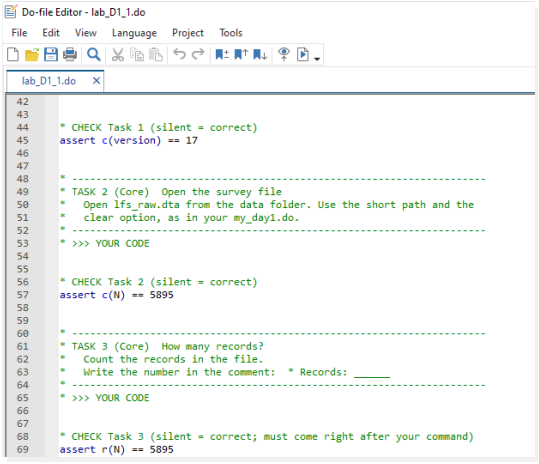
* CHECK Task 3 (silent = correct)
assert r(N) == 5895
```

Green: the task, in words

Blue: your line, in the gap

CHECK: already written. Silent means correct; red means look again.

The lab file in your editor



When your last check runs silently, put up the green note and try the Challenge task.



3

DAY 1 · PART 3

Look inside the file

Before any table, NBS checks what is really in the file.

See everything about one variable:



Type, range, number of different values, missing values, labels.

YOUR TURN · TASK 21

Open the report on labour force status

DO

1. Type `codebook ilo_lfs19`
2. Press **Enter**.

YOU SHOULD SEE

A report with the type, the number of missing values, and a small table with the labels.

IF NOT

`variable ilo_lfs19 not found`? Check `lfs19`: letters l, f, s, then 19.

EXTRA Type `codebook dem_mrt`. Which codes are not marital statuses?

Your result

```
. codebook ilo_lfs19

ilo_lfs19                                                    Labour Force Status - 19th ICLS

                  Type: Numeric (byte)
                 Label: label_ilo_lfs19

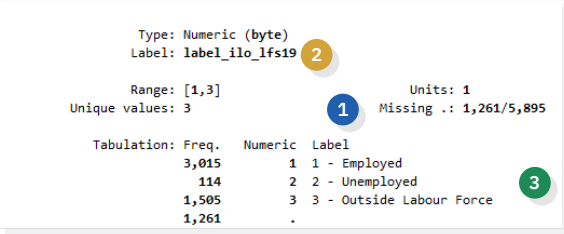
                 Range: [1,3]                          Units: 1
        Unique values: 3                           Missing .: 1,261/5,895

          Tabulation: Freq.   Numeric  Label
                      3,015         1  1 - Employed
                        114         2  2 - Unemployed
                      1,505         3  3 - Outside Labour Force
                      1,261         .
```

Read the report

```
codebook ilo_lfs19
```

1. **Missing**: no value
2. **Label**: its value label
3. **Table**: codes and counts



A second codebook: something is wrong

```
            Type: Numeric (byte)
           Label: DEM_MRT

           Range: [1,99]                        Units: 1   (1)
  Unique values: 7                          Missing .: 1,261/5,895   (2)

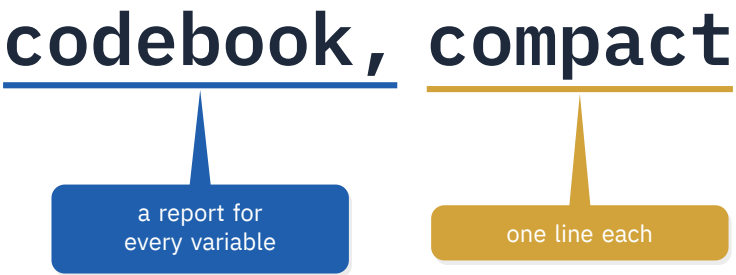
  Tabulation:  Freq.   Numeric  Label
               2,250          1  Single
               1,100          2  Married
                 987          3  Living together like married
                                 partners
                  49          4  Separated
                  77          5  Divorced
                 170          6  Widowed
                   1         99  Refusal   (3)
               1,261          .
```

1. Range [1,99]: no marital status has code 99
2. 1,261 missing: the under-15s again
3. 99 is Refusal, stored as if it were a status

Notice: the missing values in `ilo_lfs19` are the people under 15.

Labour force status is defined only for the working-age population.

The whole file in one table:



codebook, compact: one line per variable

```
Variable     Obs Unique      Mean  Min  Max  Label
-------------------------------------------------------------------------------
ilo_sex     5895      2  1.525191    1    2  Sex (2)
ilo_age     5895     86  36.70891    0   85  Age in completed years (top-coded at 85)
dem_mrt     4634      7   1.96612    1   99  Marital status (1)
ilo_lfs19   4634      3  1.674148    1    3  Labour Force Status - 19th ICLS
-------------------------------------------------------------------------------
```

1. Max 99 for marital status: worth a look
2. Obs 5895: every row has a sex and an age
3. Unique: how many different values

Name the variables you want:

`describe ilo_sex`	one variable
`describe ilo_sex ilo_age`	several: separate them with spaces
`describe dif_*`	every name that starts with `dif_`
`describe ilo_job1_*`	every main-job variable

The * stands for any letters. Names come from the questionnaire module.

YOUR TURN · TASK 22

Describe one questionnaire module

DO

1. Type `describe dif_*`
2. Press **Enter**.

YOU SHOULD SEE

The six difficulty variables: seeing, hearing, walking, remembering, self-care, communicating.

IF NOT

`variable dif not found`? Add the underscore before the `*`.

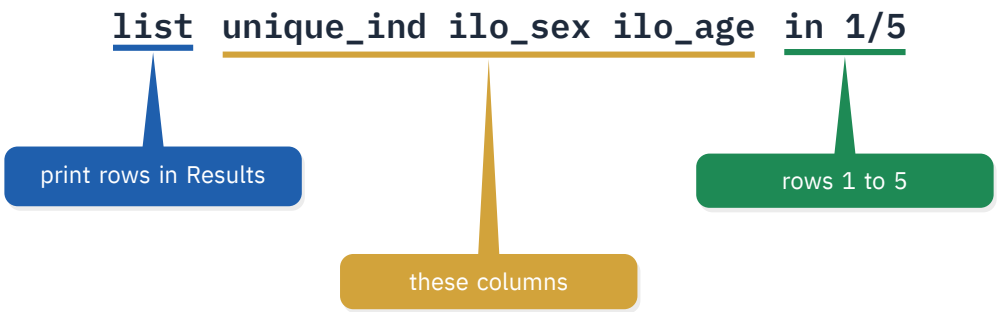
EXTRA Type `describe ilo_job1_*`. How many main-job variables?

Your result

```
. describe dif_*

Variable      Storage   Display    Value
    name         type    format    label      Variable label
------------------------------------------------------------------------------
dif_sight       byte    %24.0g     DIF_SIGHT
                                              Difficultly seeing
dif_hear        byte    %24.0g     DIF_HEAR   Difficulty hearing
dif_mobi        byte    %24.0g     DIF_MOBI   Difficulty walking
dif_conc        byte    %24.0g     DIF_CONC   Difficulty remembering
dif_care        byte    %24.0g     DIF_CARE   Difficulty with self-care
dif_comm        byte    %24.0g     DIF_COMM   Difficulty communicating
```

Show the first rows on screen:



YOUR TURN · TASK 23

List the first five records

DO

Type:

```
list unique_ind ilo_sex ilo_age in 1/5
```

and press **Enter**.

YOU SHOULD SEE

A small table: five rows, three columns, with `Male` and `Female` as words next to their codes.

IF NOT

`invalid syntax`? Check the slash in `1/5`.

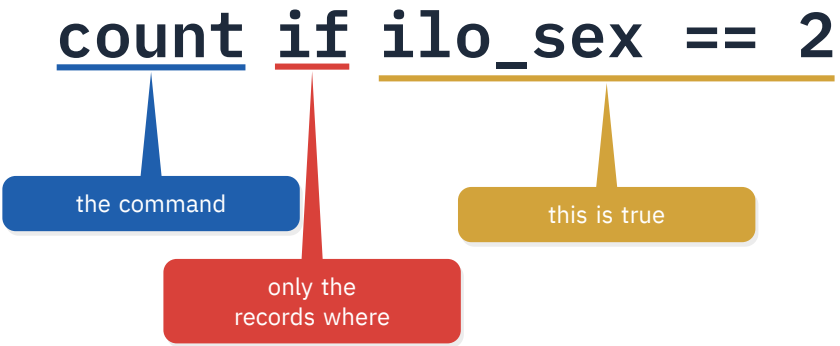
EXTRA List the same three variables `in 100/105`.

Your result

```
. list unique_ind ilo_sex ilo_age in 1/5
```

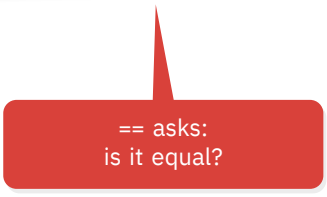
	uniqu~nd	ilo_sex	ilo_age
1.	1121101	2 - Female	60
2.	1121102	1 - Male	31
3.	1121103	1 - Male	9
4.	112111	1 - Male	47
5.	112112	2 - Female	48

Work on part of the file:



Two equals signs to compare

```
count if ilo_sex == 2
```



One = sets a value. You meet it in Part 6.

YOUR TURN · TASK 24

Count the women

DO

Type:

```
count if ilo_sex == 2
```

and press **Enter**.

YOU SHOULD SEE

One number: the same as the **Female** row in your `tabulate` table.

IF NOT

Red message after one =? Use two: ==.

EXTRA Count the men.

Your result

```
. count if ilo_sex == 2
  3,096

.
```

COMMON ERROR

One equals sign

```
. count if ilo_sex = 2
=exp not allowed
```

WHY

A single = means “make equal”. In a condition Stata needs the question “is it equal?”, written ==.

FIX

```
count if ilo_sex == 2
```

Combine conditions

`&`	and
`|`	or
`!=`	is not equal to
`<`	less than
`>=`	at least

```
count if ilo_sex == 2 & ilo_lfs19 == 2
```

Women who are unemployed (`ilo_lfs19` code 2).

YOUR TURN · TASK 25

Count unemployed women

DO

Type:

```
count if ilo_sex == 2 & ilo_lfs19 == 2
```

and press **Enter**.

YOU SHOULD SEE

One number, much smaller than the number of women.

IF NOT

Zero? Check both conditions: `ilo_sex` 2, `ilo_lfs19` 2.

EXTRA Count unemployed people in Praslin & LaDigue: `region_q == 6`.

Your result

```
. count if ilo_sex == 2 & ilo_lfs19 == 2
  59


.
```

COMMON ERROR

Half a condition

```
. count if ilo_age > 60 & < 70
<70 invalid name
```

WHY

Each side of `&` must be a full condition. Stata does not repeat `ilo_age` for you.

FIX

```
count if ilo_age > 60 & ilo_age < 70
```

ON YOUR OWN · B

Persons aged 65 and over

THE QUESTION

How many persons aged 65 and over are in the file? How many of them are women?

HINT

Two `count if` lines; the second adds `& ilo_sex == 2`.

ON YOUR OWN · B · ANSWER

```
count if ilo_age >= 65
count if ilo_age >= 65 & ilo_sex == 2
```

```
. count if ilo_age >= 65
  681 (1)

. count if ilo_age >= 65 & ilo_sex == 2
  404 (2)
```

1. 681 persons aged 65+
2. 404 of them women

RULE

In Stata, missing is larger than any number.

```
count if wkt_mj_acthrs > 60
```

also counts every record with no hours

```
count if wkt_mj_acthrs > 60 & !missing(wkt_mj_acthrs)
```

leaves out the records with no hours

The same question, two answers

```
. count if wkt_mj_acthrs > 40
  3,888 (1)

. count if wkt_mj_acthrs > 40 & !missing(wkt_mj_acthrs)
  1,011 (2)
```

1. 3,888 “work more than 40 hours”: every person with no hours is counted, because missing is larger than any number
2. 1,011: only persons who answered

QUICK CHECK

`count if wkt_mj_acthrs > 40` gives 3,888, more than all the employed. Why?

A. many people work long hours

B. missing counts as larger than any number

C. the duplicates are counted twice

ANSWER

`count if wkt_mj_acthrs > 40` gives 3,888, more than all the employed. Why?

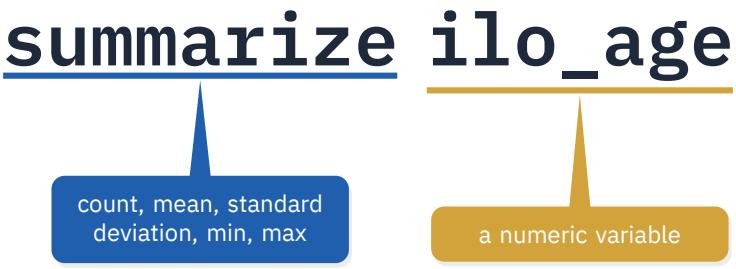
A many people work long hours

B missing counts as larger than any number

C the duplicates are counted twice

Stata stores missing as a very large number, so `> 40` is true for it. Add `& !missing(wkt_mj_acthrs)` to every `>` condition.

Describe a number in five figures:



YOUR TURN · TASK 26

Summarize age and hours

DO

Type:

```
summarize ilo_age wkt_mj_acthrs
```

and press **Enter**.

YOU SHOULD SEE

Two rows: Obs, Mean, Std. dev., Min, Max.

IF NOT

Fewer Obs for hours? That is correct: only people with a job report hours.

EXTRA Look at the Max for hours. Is it possible in one week?

Your result

```
. summarize ilo_age wkt_mj_acthrs
```

Variable	Obs	Mean	Std. dev.	Min	Max
ilo_age	5,895	36.70891	21.95336	0	85
wkt_mj_act~s	3,018	43.4672	63.98016	0	997

A number to question

```
. summarize ilo_age wkt_mj_acthrs
```

Variable	Obs	Mean	Std. dev.	Min	Max
ilo_age	5,895	36.70891	21.95336	0	85
wkt_mj_act~s	3,018	43.4672	63.98016	0	997

997 hours in one week?

Nobody works that many hours. We deal with it in Part 4.

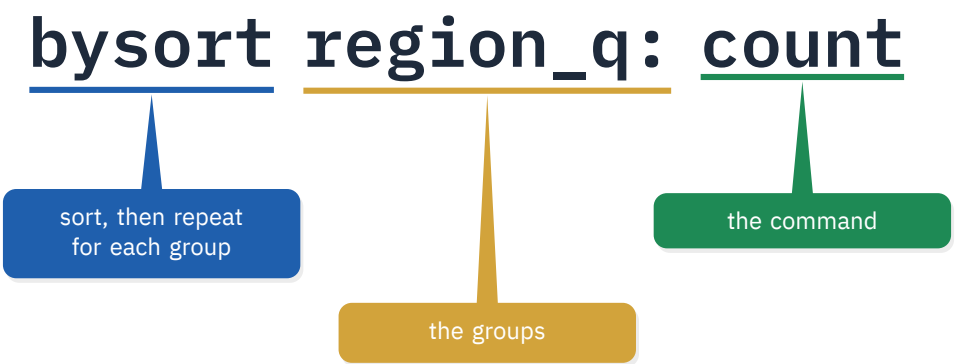
summarize ignores text

```
. summarize mig_ctz

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
     mig_ctz |          0 (1)
```

(1) Obs 0: `mig_ctz` holds letters, so there is nothing to average

The same command, group by group:



`bysort` does `sort` and `by` in one step.

YOUR TURN · TASK 27

Count records in each region

DO

Type:

```
bysort region_q: count
```

and press **Enter**.

YOU SHOULD SEE

Seven blocks, one per region, each with a count.

IF NOT

`not sorted`? You typed `by`; use `bysort`.

EXTRA Type `bysort ilo_sex: summarize ilo_age`.

Your result

```
. bysort region_q: count

-------------------------------------------------------------------------------
-> region_q = 1- North
  943

-------------------------------------------------------------------------------
-> region_q = 2- Central 1
  882

-------------------------------------------------------------------------------
-> region_q = 3- Central 2
  843

-------------------------------------------------------------------------------
-> region_q = 4- EastSouth
  1,451

-------------------------------------------------------------------------------
-> region_q = 5- West
  900

-------------------------------------------------------------------------------
-> region_q = 6- Praslin & LaDigue
  514

-------------------------------------------------------------------------------
-> region_q = 7- Perseverance
  362

.
```

ON YOUR OWN · C

Age of the unemployed

THE QUESTION

What is the average age of the unemployed (`ilo_lfs19 == 2`)? The youngest and the oldest?

HINT

`summarize` with an `if`.

ON YOUR OWN · C · ANSWER

```
summarize ilo_age if ilo_lfs19 == 2
```

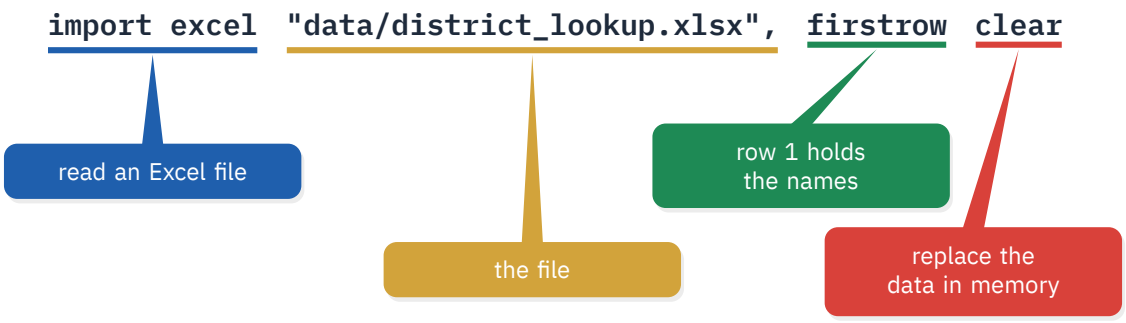
```
    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
     ilo_age |        114    30.05263    13.00091        15         63
```

1

1 114 unemployed; mean age 30.1; from 15 to 63

NBS files also arrive from Excel and from CSV exports. Stata reads both.

Open an Excel sheet:



YOUR TURN · TASK 28

Import the district list

DO

Type in your do-file, then run the line:

```
import excel "data/district_lookup.xlsx",
firstrow clear
```

Then type `browse`.

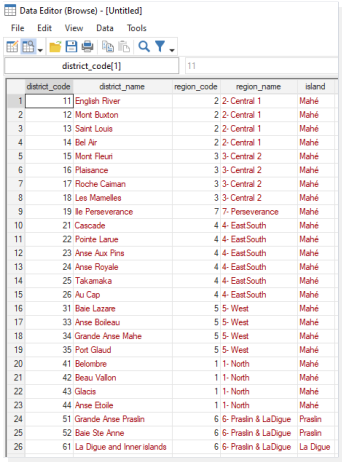
YOU SHOULD SEE

26 rows, one per district, with proper column names.

IF NOT

Columns named `A`, `B`, `C`? Add `firstrow`.

Your result


Data Editor (Browse) - [Untitled]

File Edit View Data Tools

district_code[1] | 11

	district_code	district_name	region_code	region_name	island
1	11	English River	2	2- Central 1	Mahé
2	12	Mont Buxton	2	2- Central 1	Mahé
3	13	Saint Louis	2	2- Central 1	Mahé
4	14	Bel Air	2	2- Central 1	Mahé
5	15	Mont Fleuri	3	3- Central 2	Mahé
6	16	Plaisance	3	3- Central 2	Mahé
7	17	Roche Caiman	3	3- Central 2	Mahé
8	18	Les Mamelles	3	3- Central 2	Mahé
9	19	Ile Perseverance	7	7- Perseverance	Mahé
10	21	Cascade	4	4- EastSouth	Mahé
11	22	Pointe Larue	4	4- EastSouth	Mahé
12	23	Anse Aux Pins	4	4- EastSouth	Mahé
13	24	Anse Royale	4	4- EastSouth	Mahé
14	25	Takamaka	4	4- EastSouth	Mahé
15	26	Au Cap	4	4- EastSouth	Mahé
16	31	Baie Lazare	5	5- West	Mahé
17	33	Anse Boileau	5	5- West	Mahé
18	34	Grande Anse Mahe	5	5- West	Mahé
19	35	Port Glaud	5	5- West	Mahé
20	41	Belombre	1	1- North	Mahé
21	42	Beau Vallon	1	1- North	Mahé
22	43	Glacis	1	1- North	Mahé
23	44	Anse Etoile	1	1- North	Mahé
24	51	Grande Anse Praslin	6	6- Praslin & LaDigue	Praslin
25	52	Baie Ste Anne	6	6- Praslin & LaDigue	Praslin
26	61	La Digue and Inner islands	6	6- Praslin & LaDigue	La Digue

Why `firstrow` matters

WITHOUT FIRSTROW

```
. import excel "data/district_lookup.xlsx", clear
(5 vars, 27 obs)
     +-------------------------------------------------------------------+
     |            A              B            C             D        E |
     |-------------------------------------------------------------------|
  1. | district_code   district_name   region_code   region_name   island |
  2. |            11   English River             2   2- Central 1   Mahé |
  3. |            12     Mont Buxton             2   2- Central 1   Mahé |
```

WITH FIRSTROW

```
. import excel "data/district_lookup.xlsx", firstrow clear
(5 vars, 26 obs)
     +-----------------------------------------------------------------+
     | distr~de   district_name   regio~de   region_name   island |
     |-----------------------------------------------------------------|
  1. |       11   English River           2   2- Central 1     Mahé |
  2. |       12     Mont Buxton           2   2- Central 1     Mahé |
  3. |       13     Saint Louis           2   2- Central 1     Mahé |
```

Without `firstrow`: columns A to E and 27 rows, the names became data. With it: 26 districts and real names.

QUICK CHECK

An Excel sheet of 26 districts is imported and Stata reports 27 observations. Why?

A. Excel adds an empty row

B. the header row was read as data: firstrow is missing

C. one district is duplicated

ANSWER

An Excel sheet of 26 districts is imported and Stata reports 27 observations. Why?

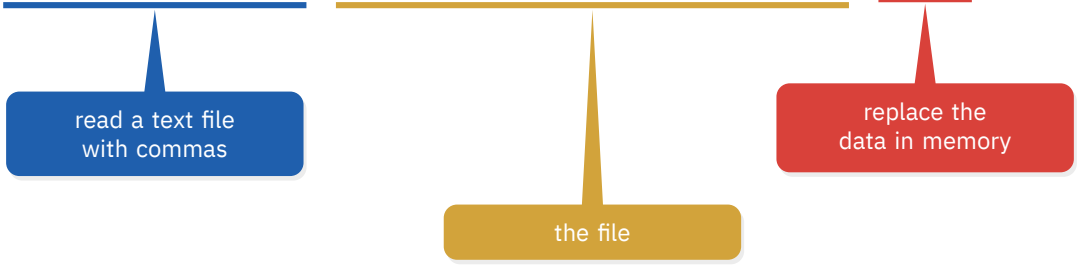
A. Excel adds an empty row

B. the header row was read as data: firstrow is missing

C. one district is duplicated

Without `firstrow` the column names become the first observation, and every column is read as text.

Open a CSV file:



YOUR TURN · TASK 29

Import the same list from CSV

DO

Type and run:

```
import delimited
"data/district_lookup.csv", clear
```

Then `describe`.

YOU SHOULD SEE

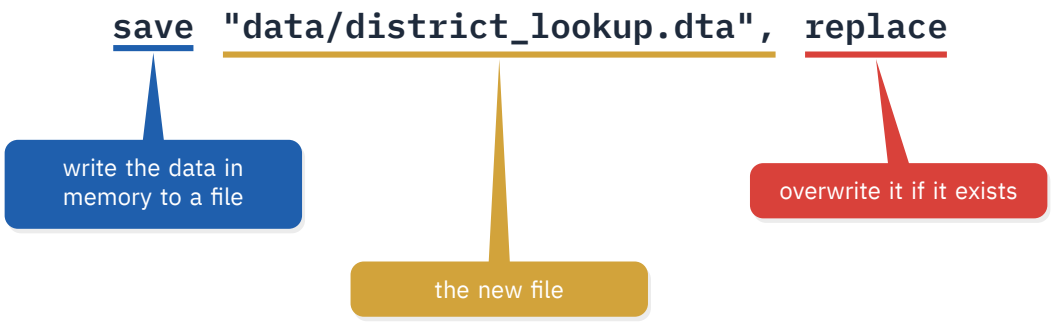
26 observations and the same columns as from Excel.

IF NOT

`file not found`? Check the `cd` line ran first.

EXTRA Compare the storage types with the Excel version.

Keep it as a Stata file:



YOUR TURN · TASK 30

Save the list, go back to the survey

DO

Add and run:

```
save "data/district_lookup.dta", replace
use "data/lfs_raw.dta", clear
```

YOU SHOULD SEE

`file data/district_lookup.dta` saved, then the survey file back in the Variables window.

IF NOT

`file already exists?` Add `, replace`.

EXTRA Find `district_lookup.dta` in your `data` folder.

COMMON ERROR

Importing over changed data

```
. import excel "data/district_lookup.xlsx", firstrow
no; data in memory would be lost
```

WHY

Same rule as `use`: memory holds changes, and `import` would replace them.

FIX

Add `clear`: `import excel "data/district_lookup.xlsx", firstrow clear`

RULE

`import` and `use` replace the data in memory.

Stata holds one file at a time. Save your work before you open another file.

Part 3 on one page

CHEAT CARD · KEEP THIS ONE

`codebook v`	type, range, missing, the codes
`codebook v1 v2, compact`	one line per variable
`describe dif_*`	every variable starting dif_
`list v1 v2 in 1/5`	the first five rows
`count if v == 2`	how many rows meet a condition
`& | != < >=`	and, or, not equal, less, at least
`... & !missing(v)`	keep missing out of > conditions
`summarize v`	obs, mean, sd, min, max
`bysort g: summarize v`	the same, group by group
`import excel "f.xlsx", firstrow clear`	read an Excel sheet
`import delimited "f.csv", clear`	read a CSV file
`save "data/f.dta", replace`	save the data in memory

Practice on your own

PRACTICE SET · practice_D1.do, PART 3

CORE	P1	How many records have `ilo_lfs19` missing? Use `count if`.
CORE	P2	List `unique_ind ilo_age dem_mrt` for records 101 to 105.
CORE	P3	Mean age of men and of women, with `bysort`.
EXTENSION	P4	How many employed persons work more than 48 actual hours? Watch the don't-know code.
CHALLENGE	P5	Import `district_lookup.csv` and count the districts on Mahé.

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

You can now inspect any NBS file, select groups with `if`, and bring in Excel and CSV files.

lost? open **catchup_D1_S3.do**



4

DAY 1 · PART 4

Don’t know is not a number

Three kinds of “no answer”

Not asked

the questionnaire skipped it

.

Don’t know

the respondent could not say

97

Refused

the respondent would not say

99

The codes 97 and 99 are examples. Each variable’s value label says which code means what.

In this file, 97 and 99 are stored as numbers. A mean would treat them as real ages or hours.

YOUR TURN · TASK 31

Find the don’t-know codes in marital status

DO

Type:

```
tabulate dem_mrt, missing
```

and press **Enter**.

YOU SHOULD SEE

A table of marital status. Near the bottom, a **Refusal** row, then a row for `.`

IF NOT

`option missing not allowed`? Put the comma before `missing`.

EXTRA Add `nolabel` after `missing`: which code is behind Refusal?

Your result

```
. tabulate dem_mrt, missing
```

Marital status	Freq.	Percent	Cum.
Single	2,250	38.17	38.17
Married	1,100	18.66	56.83
Living together like married partners	987	16.74	73.57
Separated	49	0.83	74.40
Divorced	77	1.31	75.71
Widowed	170	2.88	78.59
Refusal	1	0.02	78.61
.	1,261	21.39	100.00
Total	5,895	100.00	

Two kinds of no answer in one table

Marital status	Freq.	Percent	Cum.	
Single	2,250	38.17	38.17	
Married	1,100	18.66	56.83	
Living together like married partners	987	16.74	73.57	
Separated	49	0.83	74.40	
Divorced	77	1.31	75.71	
Widowed	170	2.88	78.59	
Refusal	1	0.02	78.61	1
.	1,261	21.39	100.00	2
Total	5,895	100.00		

1. Refusal: 1 person refused, stored as code 99
2. . : 1,261 never asked (under 15)
3. both are “no answer”, for different reasons: keep the reason

Two kinds of missing in Stata



system missing

no value, no reason

. a . b … . z

extended missing

no value, **reason kept**

Both are left out of means, totals and tables. Only the second remembers why.

Turn codes into missing, with the reason:

```
mvdecode dem_mrt dem_edl, mv(97=.a \ 99=.b)
```

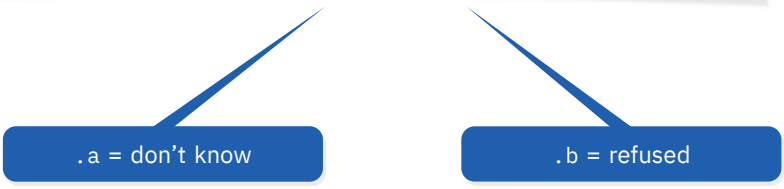
codes to missing

these variables

97 becomes .a,
99 becomes .b

The letter keeps the reason

```
mvdecode dem_mrt dem_edl, mv(97=.a \ 99=.b)
```



Write the meaning of each letter in a comment above the line.

YOUR TURN · TASK 32

Recode marital status and education

DO

Add to `my_day1.do` and run:

```
* .a = don't know, .b = refused
mvdecode dem_mrt dem_edl, mv(97=.a \ 99=.b)
tabulate dem_mrt, missing
```

YOU SHOULD SEE

A line saying how many values were made missing, then the table with `.a` or `.b` rows instead of the codes.

IF NOT

`invalid syntax`? Check the backslash `\` between the two rules.

Your result

```
. mvdecode dem_mrt dem_edl, mv(97=.a \ 99=.b)
     dem_mrt: 1 missing value generated
     dem_edl: 134 missing values generated

. tabulate dem_mrt, missing
```

Marital status	Freq.	Percent	Cum.
Single	2,250	38.17	38.17
Married	1,100	18.66	56.83
Living together like married partners	987	16.74	73.57
Separated	49	0.83	74.40
Divorced	77	1.31	75.71
Widowed	170	2.88	78.59
.	1,261	21.39	99.98
.b	1	0.02	100.00
Total	5,895	100.00	

What mvdecode reports

```
. mvdecode dem_mrt dem_edl, mv(97=.a \ 99=.b)
     dem_mrt: 1 missing value generated
     dem_edl: 134 missing values generated

. tabulate dem_mrt, missing

                        Marital status |      Freq.     Percent        Cum.
---------------------------------------+-----------------------------------
                                Single |      2,250       38.17       38.17
                               Married |      1,100       18.66       56.83
Living together like married partners |        987       16.74       73.57
                             Separated |         49        0.83       74.40
                              Divorced |         77        1.31       75.71
                               Widowed |        170        2.88       78.59
                                     . |      1,261       21.39       99.98
                                    .b |          1        0.02      100.00
---------------------------------------+-----------------------------------
                                 Total |      5,895      100.00
```

1. dem_mrt: 1 value turned into a missing
2. dem_edl: 134
3. the refusal is now `.b`: missing, with its reason

RULE

Read the label before you recode.

In one variable 97 means don’t know. In another it can be a real answer.

`codebook varname` first, then `mvdecode`

YOUR TURN · TASK 33

Check first, then recode

DO

1. Type `codebook mie_socpro`.
2. Is 97 labelled as don't know?
3. If yes, add to your do-file:
   `mvdecode mie_socpro, mv(97=.a)`

YOU SHOULD SEE

The codebook shows the code and its label; after the recode, `.a` in its place.

IF NOT

No 97 in the codebook? Then nothing to recode: that is a correct answer.

EXTRA Find two more variables that use 97. Hint: `codebook, compact`.

Your result

```
. codebook mie_socpro

-------------------------------------------------------------------------------
mie_socpro                                        Employer pays social contributions
-------------------------------------------------------------------------------

                  Type: Numeric (byte)
                 Label: MIE_SOCPRO

                 Range: [1,97]                        Units: 1
         Unique values: 3                         Missing .: 3,330/5,895

            Tabulation: Freq.   Numeric  Label
                        2,245         1  YES
                          280         2  NO
                           40        97  DON'T KNOW
                        3,330         .
```

Read the label before you recode

```
                  Type: Numeric (byte)
                 Label: MIE_SOCPRO

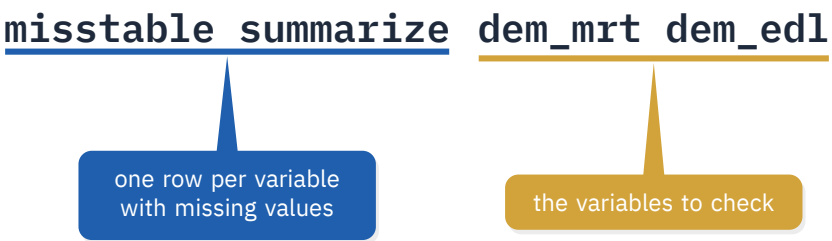
                 Range: [1,97]                        Units: 1
         Unique values: 3                         Missing .: 3,330/5,895

            Tabulation: Freq.   Numeric  Label
                        2,245         1  YES
                          280         2  NO
                           40        97  DON'T KNOW
                        3,330         .

. mvdecode mie_socpro, mv(97=.a)
  mie_socpro: 40 missing values generated
```

1. 97 means DON'T KNOW in this variable: 40 persons
2. after `mvdecode`: 40 missing values generated

See the missing values of many variables:



YOUR TURN · TASK 34

Map the missing values

DO

Type:

```
misstable summarize dem_mrt dem_edl
ilo_lfs19 wkt_mj_acthrs
```

and press **Enter**.

YOU SHOULD SEE

One row per variable with columns for `.`, for `.a` to `.z`, and for the non-missing values.

IF NOT

A variable is not in the table? It has no missing values.

EXTRA Run it on `dif_*`.

Your result

```
. misstable summarize dem_mrt dem_edl ilo_lfs19 wkt_mj_acthrs
```

Variable	Obs=.	Obs>.	Obs<.	Obs<. Unique values	Obs<. Min	Obs<. Max
dem_mrt	1,261	1	4,633	6	1	6
dem_edl	353	134	5,408	18	1	18
ilo_lfs19	1,261		4,634	3	1	3
wkt_mj_act~s	2,877		3,018	80	0	997

Read misstable

Variable	Obs=.	Obs>.	Obs<.	Obs<. Unique values	Obs<. Min	Obs<. Max
dem_mrt	1,261	1	4,633	6	1	6
dem_edl	353	134	5,408	18	1	18
dem_edc	3,896		1,999	2	1	2
ilo_lfs19	1,261		4,634	3	1	3

1. Obs=. : plain missing (not asked)
2. Obs>. : the coded missings, .a and .b
3. dem_edc: 3,896 not asked: a question for some ages only

Many missing values are correct

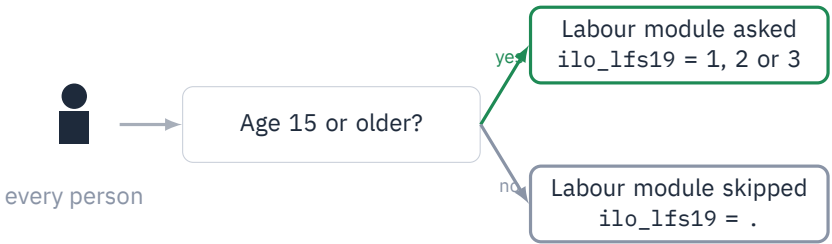
```
. misstable summarize dem_mrt dem_edl ilo_lfs19 wkt_mj_acthrs
```

Variable	Obs=.	Obs>.	Obs<.	Obs<. Unique values	Obs<. Min	Obs<. Max
dem_mrt	1,261	1	4,633	6	1	6
dem_edl	353	134	5,408	18	1	18
ilo_lfs19	1,261		4,634	3	1	3
wkt_mj_act~s	2,877		3,018	80	0	997

questions not asked:
a correct .

Under-15s have no labour status. People without a job have no hours.

The questionnaire’s routing, drawn



YOUR TURN · TASK 35

Let Stata confirm the routing

DO

Add to your do-file and run:

```
assert missing(ilo_lfs19) if ilo_age < 15
```

YOU SHOULD SEE

Nothing. Silence means the statement is true for every record.

IF NOT

Red `assertion is false`? Some under-15s have a labour status: tell the trainer.

EXTRA Write the opposite check: `assert !missing(ilo_lfs19) if ilo_age >= 15`.

assert: silent when true

```
assert missing(ilo_lfs19) if ilo_age < 15
```

says nothing when true; stops the do-file when false

A failed assert stops the run before a wrong table is produced.

COMMON ERROR

An assert that fails tells you how often

```
. assert ilo_age >= 15
1,261 contradictions in 5,895 observations
assertion is false
```

WHY

The claim “everyone is at least 15” is false for 1,261 rows: the children. A failed assert stops the do-file.

FIX

Assert only what must be true: `assert missing(ilo_lfs19) if ilo_age < 15` is silent.

Hours, before and after the code is removed

```
     Actual |
   hours in |
   main job |      Freq.     Percent        Cum.
------------+-----------------------------------
 DON'T KNOW |         13      100.00      100.00
------------+-----------------------------------
      Total |         13      100.00
. mvdecode wkt_mj_acthrs, mv(997=.a)

wkt_mj_act~s: 13 missing values generated
    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
wkt_mj_act~s |      3,005     39.3421    12.62638          0        168
```

1. every value above 168 is DON'T KNOW: 13 persons
2. recoded to .a
3. mean 39.3 hours, max 168: before, 43.5 and 997

QUICK CHECK

Mean actual hours fell from 43.5 to 39.3. What changed in the data?

A. 13 don’t-know codes are no longer counted as 997 hours

B. the duplicates were removed

C. the under-15s were dropped

ANSWER

Mean actual hours fell from 43.5 to 39.3. What changed in the data?

A 13 don’t-know codes are no longer counted as 997 hours

B the duplicates were removed

C the under-15s were dropped

13 records had 997, the code for don’t know. As numbers they added almost 13,000 hours. As .a they are left out of the mean.

Part 4 on one page

CHEAT CARD · KEEP THIS ONE

`tabulate v, missing`	show the missing rows
`mvdecode v1 v2, mv(97=.a \ 99=.b)`	codes become missing, with a reason
`codebook v`	read the code labels first
`misstable summarize v1 v2`	how much is missing, of which kind
`assert condition`	stop if a rule is broken
`missing(v) / !missing(v)`	is missing / is not missing

Practice on your own

PRACTICE SET · practice_D1.do, PART 4

CORE	P1	Recode the don’t-know code of `mie_pvaclv` after reading its label.
CORE	P2	`misstable summarize` for `mie_*`: which variable has the most `.a`?
CORE	P3	Assert that `ilo_age` is never above 85.
EXTENSION	P4	Find the don’t-know code of usual hours and recode it.
CHALLENGE	P5	Write one assert that checks: anyone with a status is at least 15.

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

A production statistician writes down what must be true, and lets Stata check it.

Don’t-know codes now carry their reason and cannot distort a mean.

lost? open `catchup_D1_S4.do`



5

DAY 1 · PART 5

Tables a reader understands

The same table, twice

without labels

Sex	Freq.
1	2,799
2	3,096

Which is which?

with labels

Sex	Freq.
Male	2,799
Female	3,096

Read at a glance.

YOUR TURN · TASK 36

Find variables with no label

DO

Type:

```
describe weight dis_weight weights_dis
```

and press **Enter**.

YOU SHOULD SEE

Three rows. Look at the last column, **Variable label**.

IF NOT

`variable weight not found`? The raw file keeps NBS names: check the spelling.

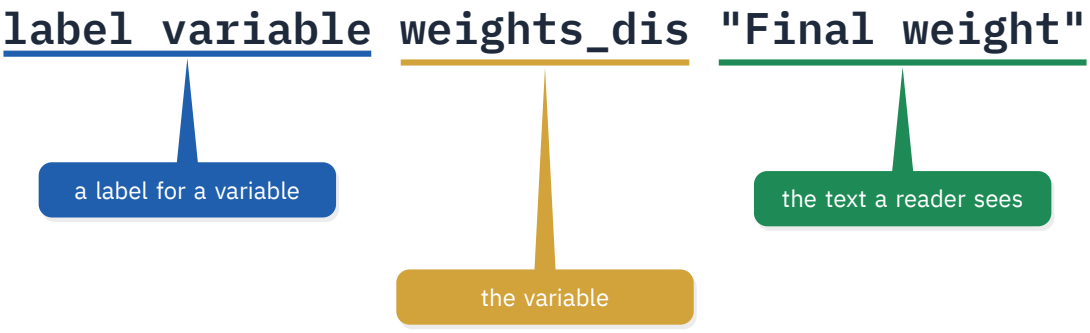
Your result

```
. describe weight dis_weight weights_dis

Variable      Storage   Display    Value
    name         type    format    label      Variable label
------------------------------------------------------------------------------
weight          float   %9.0g
dis_weight      float   %9.0g
weights_dis     float   %9.0g

.
```

Name a variable for the reader:



YOUR TURN · TASK 37

Label the three weights

DO

Add to `my_day1.do` and run:

```
label variable weight "Base weight"
label variable dis_weight "District adjustment"
label variable weights_dis "Final weight"

describe weight dis_weight weights_dis
```

YOU SHOULD SEE

The same three rows, now with their labels.

IF NOT

Red text? Check the quotes: one at the start, one at the end.

EXTRA Label `int_month` with `"Interview month"`.

Your result

```
. describe weight dis_weight weights_dis

Variable      Storage   Display    Value
    name         type    format    label      Variable label
------------------------------------------------------------------------------
weight          float   %9.0g                 Base weight
dis_weight      float   %9.0g                 District adjustment
weights_dis     float   %9.0g                 Final weight

.
```

Before and after `label variable`

BEFORE

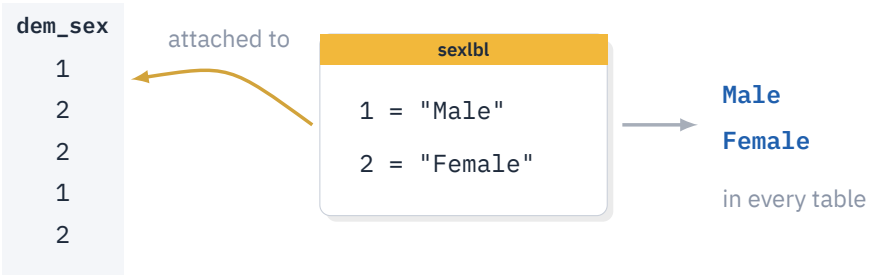
```
Variable      Storage   Display    Value
    name         type    format    label      Variable label
------------------------------------------------------------
weight          float   %9.0g
dis_weight      float   %9.0g
weights_dis     float   %9.0g
```

AFTER

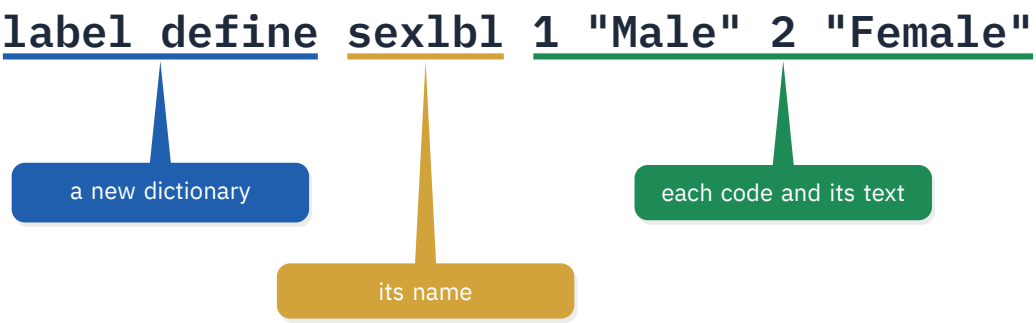
```
Variable      Storage   Display    Value
    name         type    format    label      Variable label
------------------------------------------------------------
weight          float   %9.0g                 Base weight
dis_weight      float   %9.0g                 District adjustment
weights_dis     float   %9.0g                 Final weight
```

Three weights with no label: nobody knows which one to use. Three words fix it for everyone who opens the file.

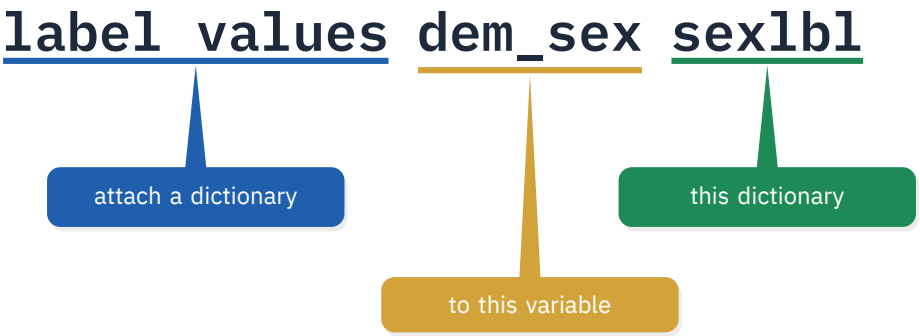
A value label is a dictionary



Step 1. Write the dictionary:



Step 2. Attach it to a variable:



Read a dictionary:



The dictionary's name is in the **Value label** column of `describe`.

YOUR TURN · TASK 38

Give sex readable labels

DO

1. Check the codes: `label list DEM_SEX`
2. If 1 is MALE and 2 is FEMALE, add and run:

```
label define sexlbl 1 "Male" 2 "Female"
label values dem_sex sexlbl
```

```
tabulate dem_sex
```

YOU SHOULD SEE

The table shows `Male` and `Female` instead of capitals.

IF NOT

`label sexlbl already defined`? Add `, replace` to the define line.

Before and after

```
. label list DEM_SEX
DEM_SEX:
           1 MALE
           2 FEMALE

. tabulate dem_sex
```

Sex	Freq.	Percent	Cum.
MALE	2,799	47.48	47.48
FEMALE	3,096	52.52	100.00
Total	5,895	100.00	

```
. label define sexlbl 1 "Male" 2 "Female"

. label values dem_sex sexlbl

. tabulate dem_sex
```

Sex	Freq.	Percent	Cum.
Male	2,799	47.48	47.48
Female	3,096	52.52	100.00
Total	5,895	100.00	

From capitals to a clean label

```
. label list DEM_SEX
DEM_SEX:
           1 MALE
           2 FEMALE

. label define sexlbl 1 "Male" 2 "Female"

. label values dem_sex sexlbl

. tabulate dem_sex

        Sex |      Freq.     Percent        Cum.
------------+-----------------------------------
       Male |      2,799       47.48       47.48
     Female |      3,096       52.52      100.00
------------+-----------------------------------
      Total |      5,895      100.00
```

1. the NBS label: MALE, FEMALE in capitals
2. attach the new dictionary
3. the table now reads Male, Female

COMMON ERROR

Defining a label twice

```
. label define sexlbl 1 "Male" 2 "Female"
label sexlbl already defined
```

WHY

A label name can be defined once. Running the do-file a second time defines it again.

FIX

Add `, replace`: `label define sexlbl 1 "Male" 2 "Female", replace`

QUICK CHECK

You ran `label define yesno 0 "No" 1 "Yes"`, but the table of `employed` still shows 0 and 1. What is missing?

A. `label variable`

B. `label values employed yesno`

C. `label list`

ANSWER

You ran `label define yesno 0 "No" 1 "Yes"`, but the table of `employed` still shows 0 and 1. What is missing?

A `label variable`

B `label values employed yesno`

C `label list`

`label define` writes the dictionary; `label values` attaches it to a variable. Without the second step nothing changes on screen.

Change a dictionary that exists

```
label define sexlbl 1 "Men" 2 "Women", replace
label define sexlbl 1 "Men", modify
```

start the dictionary again

change only the codes listed

Labels for the reasons, too

1. Find the dictionary's name: `describe dem_mrt` → **`DEM_MRT`**

2. Add the two reasons to it:

```
label define DEM_MRT .a "Don't know" .b "Refusal", add
```

keep the existing codes, add new ones

YOUR TURN · TASK 39

Label the reasons in marital status

DO

Add and run:

```
label define DEM_MRT .a "Don't know" .b "Refusal", add
tabulate dem_mrt, missing
```

YOU SHOULD SEE

The `.b` row now reads **Refusal**.

IF NOT

`invalid attempt to modify label`? The code already has a text: use `modify` instead of `add`.

EXTRA Do the same for `dem_edl`: its dictionary is `DEM_EDL`.

Your result

```
. tabulate dem_mrt, missing
```

Marital status	Freq.	Percent	Cum.
Single	2,250	38.17	38.17
Married	1,100	18.66	56.83
Living together like married partners	987	16.74	73.57
Separated	49	0.83	74.40
Divorced	77	1.31	75.71
Widowed	170	2.88	78.59
.	1,261	21.39	99.98
Refusal	1	0.02	100.00
Total	5,895	100.00	

The refusal, with its name

```
. tabulate dem_mrt, missing

                       Marital status |      Freq.     Percent        Cum.
--------------------------------------+-----------------------------------
                               Single |      2,250       38.17       38.17
                              Married |      1,100       18.66       56.83
Living together like married partners |        987       16.74       73.57
                            Separated |         49        0.83       74.40
                             Divorced |         77        1.31       75.71
                              Widowed |        170        2.88       78.59
                                    . |      1,261       21.39       99.98
                              Refusal |          1        0.02      100.00
--------------------------------------+-----------------------------------
                                Total |      5,895      100.00
```

1

1 .b now reads Refusal: missing for analysis, readable for people

Part 5 on one page

CHEAT CARD · KEEP THIS ONE

`label variable v "text"`	what the variable means
`label define name 1 "A" 2 "B"`	write a dictionary of codes
`label define name ..., replace / modify`	redo it / add to it
`label define name .a "Don't know", add`	name the missing codes
`label values v name`	attach the dictionary
`label list name`	show a dictionary
`label data "text"`	describe the whole file

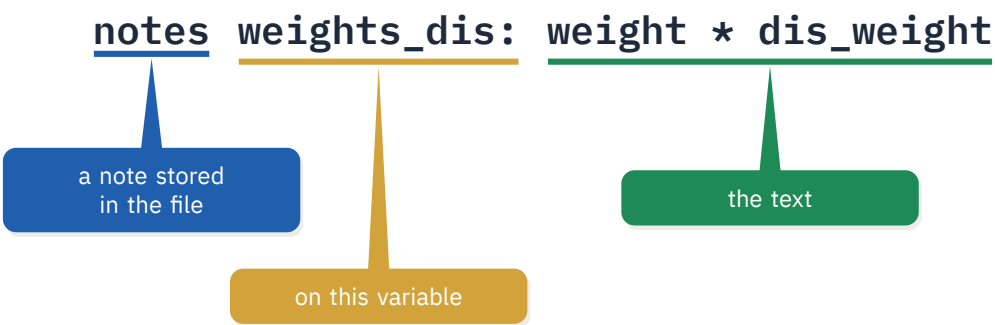
A label for the whole file:

```
label data "NBS LFS Q1 2026, teaching file"
```

describe the file itself

shown when the file opens

Note how a variable was built:



Every variable you will publish from now has a name a reader understands.

lost? open **catchup_D1_S5.do**

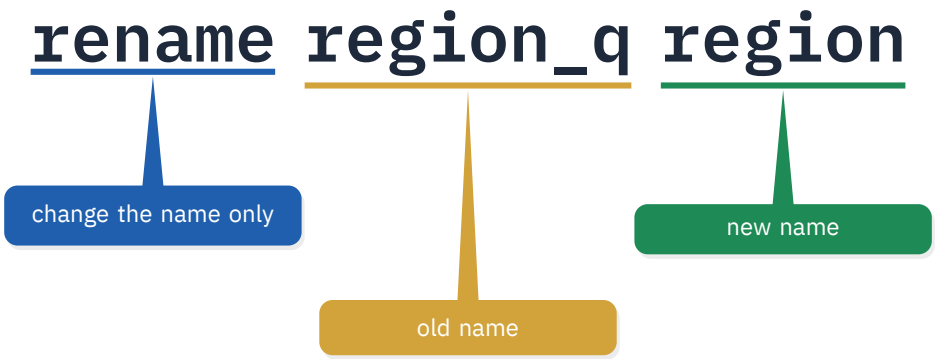


6

DAY 1 · PART 6

From answers to indicators

Give a variable a shorter name:



Values, labels and position stay the same.

YOUR TURN · TASK 40

Rename the design variables

DO

Add to `my_day1.do` and run:

```
rename region_q region
rename district_q district
rename psu_nbs_2026 psu
rename ea_q ea
```

YOU SHOULD SEE

The four new names in the Variables window.

IF NOT

`variable region_q not found`? It is already renamed: the line ran twice. Rerun the whole file from the top.

YOUR TURN · TASK 41

Rename the weights

DO

Add and run:

```
rename weight wgt_base
rename dis_weight wgt_adj
rename weights_dis wgt_final
```

YOU SHOULD SEE

`wgt_base`, `wgt_adj` and `wgt_final`, next to each other in the Variables window, with the labels from Part 5.

IF NOT

Red text? See the previous slide: rerun the whole file.

EXTRA Type `describe wgt_*`.

Your result

```
. rename region_q region

. rename district_q district

. rename psu_nbs_2026 psu

. rename ea_q ea

. rename weight wgt_base

. rename dis_weight wgt_adj

. rename weights_dis wgt_final
```

Variables

Filter variables here

Name	Label
region	Region
district	DISTRICT
psu	Primary Sampling Unit- ...
ea	ENUMERATION AREA
dem_name	Name
unique_ind	Unique Individual Level ...
unique_hhd	Unique Individual Hous...
ppno	Person number
dem_sex	Sex

Rename keeps everything else

```
Variable      Storage   Display    Value
    name         type    format    label      Variable label
------------------------------------------------------------------------------------------
region           byte    %20.0g     region_q   Region (1)
district         byte    %26.0g     DISTRICT_Q (2)
                                               DISTRICT
psu              double  %16.0f                Primary Sampling Unit- PSU: Combination of EAs
ea               int     %8.0g                 ENUMERATION AREA
wgt_base         float   %9.0g                 Base weight (3)
wgt_adj          float   %9.0g                 District adjustment
wgt_final        float   %9.0g                 Final weight
```

1. new name, same value label (`region_q`)
2. labels from NBS stay as they were
3. your Part 5 labels travel with the variable

COMMON ERROR

A name that is already taken

```
. rename district region
variable region already defined
```

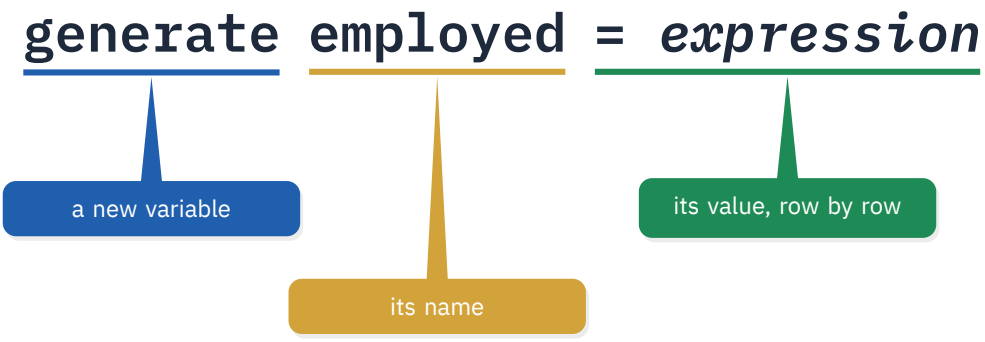
WHY

`region` exists after Task 40. Two variables cannot share a name.

FIX

Pick another name, or drop the old one first if it is truly unwanted.

Make a new variable:



One = sets a value: the one you met in Part 3.

YOUR TURN · TASK 42

Create an employed indicator

DO

Add and run:

```
generate employed = (ilo_lfs19 == 1) if !missing(ilo_lfs19)
tabulate employed ilo_lfs19, missing
```

YOU SHOULD SEE

`(1,261 missing values generated)`, then a cross-table: 1 only for Employed, 0 for the other two, `.` for the under-15s.

IF NOT

`variable employed already defined`? Rerun the whole file from the top.

A test gives 1 or 0

```
generate employed = (ilo_lfs19 == 1) if !missing(ilo_lfs19)
```



Without the `if`, every under-15 would count as not employed.

Your result

```
. generate employed = (ilo_lfs19 == 1) if !missing(ilo_lfs19)
(1,261 missing values generated)

. tabulate employed ilo_lfs19, missing
```

employed	Labour Force Status - 19th ICLS				Total
	1 - Emplo	2 - Unemp	3 - Outsi	.	
0	0	114	1,505	0	1,619
1	3,015	0	0	0	3,015
.	0	0	0	1,261	1,261
Total	3,015	114	1,505	1,261	5,895

Why the `if` is there

WITHOUT IF

```
. generate emp_noif = (ilo_lfs19 == 1)
   emp_noif |      Freq.     Percent        Cum.
------------+-----------------------------------
          0 |      2,880       48.85       48.85   1
          1 |      3,015       51.15      100.00
------------+-----------------------------------
      Total |      5,895      100.00
```

WITH IF

```
. generate employed = (ilo_lfs19 == 1) if !missing(ilo_lfs19)
   employed |      Freq.     Percent        Cum.
------------+-----------------------------------
          0 |      1,619       27.46       27.46   2
          1 |      3,015       51.15       78.61
          . |      1,261       21.39      100.00   3
------------+-----------------------------------
      Total |      5,895      100.00
```

Without the `if`, 1,261 children are counted as “not employed” (2,880 zeros). With it they stay missing.

QUICK CHECK

With `generate employed = (ilo_lfs19 == 1)` and no `if`, what does a 9-year-old get?

A. missing

B. 0: not employed

C. 1: employed

ANSWER

With `generate employed = (ilo_lfs19 == 1)` and no `if`, what does a 9-year-old get?

A missing

B 0: not employed

C 1: employed

Missing is not equal to 1, so the comparison is false: 0. The child joins the denominator of every rate unless the `if` keeps it out.

COMMON ERROR

Generate over an existing variable

```
. generate employed = (ilo_lfs19 == 1)
variable employed already defined
```

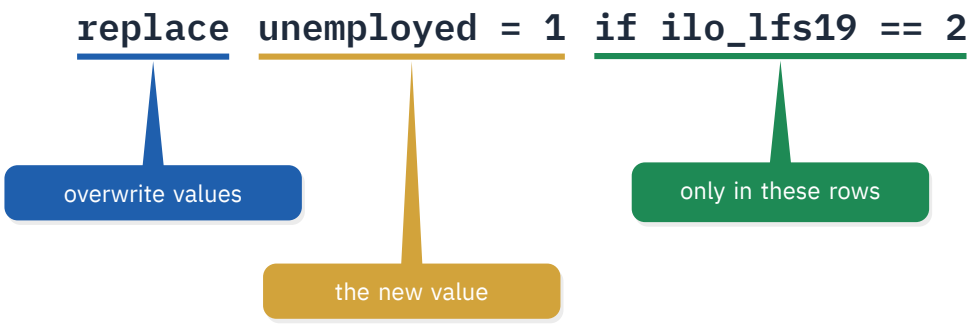
WHY

`generate` only creates. It never overwrites, so your data are safe from a typing slip.

FIX

To change values, use `replace`. To start again, `drop employed` first.

Change values that already exist:



YOUR TURN · TASK 43

Create an unemployed indicator, in two steps

DO

Add and run:

```
generate unemployed = 0 if !missing(ilo_lfs19)
replace unemployed = 1 if ilo_lfs19 == 2
tabulate unemployed ilo_lfs19, missing
```

YOU SHOULD SEE

`(114 real changes made)`, then 1 only in the Unemployed column.

IF NOT

A different number of changes? Check == 2 in the replace line.

EXTRA Create `in_lf`: 1 if employed or unemployed. Use `|`.

Your result

```
. generate unemployed = 0 if !missing(ilo_lfs19)
(1,261 missing values generated)

. replace unemployed = 1 if ilo_lfs19 == 2
(114 real changes made)

. tabulate unemployed ilo_lfs19, missing
```

unemployed	Labour Force Status - 19th ICLS 1 - Emplo	2 - Unemp	3 - Outsi	.	Total
0	3,015	0	1,505	0	4,520
1	0	114	0	0	114
.	0	0	0	1,261	1,261
Total	3,015	114	1,505	1,261	5,895

generate, then replace

```
. generate unemployed = 0 if !missing(ilo_lfs19)
(1,261 missing values generated) 1

. replace unemployed = 1 if ilo_lfs19 == 2
(114 real changes made) 2

. tabulate unemployed, missing

 unemployed |      Freq.     Percent        Cum.
------------+-----------------------------------
          0 |      4,520       76.68       76.68
          1 |        114        1.93       78.61 3
          . |      1,261       21.39      100.00
------------+-----------------------------------
      Total |      5,895      100.00
```

1. step 1: everyone with a status gets 0
2. step 2: the 114 unemployed become 1
3. children stay missing

ON YOUR OWN · D

In the labour force

THE QUESTION

Create `in_lf`: 1 for employed or unemployed, 0 for outside the labour force, missing for children. How many are in the labour force?

HINT

One `generate` with `|` (or) and an `if`.

ON YOUR OWN · D · ANSWER

```
generate in_lf = (ilo_lfs19 == 1 | ilo_lfs19 == 2)
///
if !missing(ilo_lfs19)
tabulate in_lf, missing
```

```
. tabulate in_lf, missing

      in_lf |      Freq.     Percent        Cum.
------------+-----------------------------------
          0 |      1,505       25.53       25.53
          1 |      3,129       53.08       78.61   1
          . |      1,261       21.39      100.00
------------+-----------------------------------
      Total |      5,895      100.00
```

1. 3,129 in the labour force = 3,015 employed + 114 unemployed

YOUR TURN · TASK 44

Check that nobody is both

DO

Add and run:

```
assert employed + unemployed <= 1 if
!missing(ilo_lfs19)
```

YOU SHOULD SEE

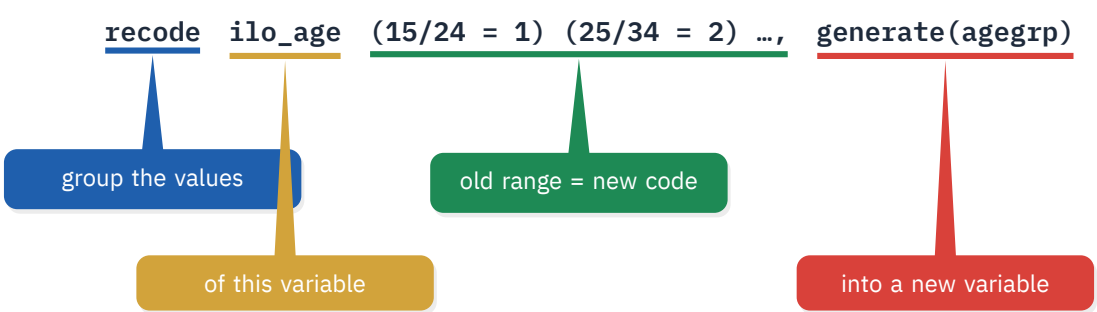
Nothing. Silence means no record is employed and unemployed at once.

IF NOT

`assertion is false`? One of the two generate lines is wrong: compare with the slides.

EXTRA Write an assert that `unemployed` is never 1 for the under-15s.

Group values into classes:



YOUR TURN · TASK 45

Make the age groups of the release

DO

Add and run:

```
recode ilo_age (0/14 = .) ///
  (15/24 = 1) (25/34 = 2) ///
  (35/54 = 3) (55/max = 4), ///
  generate(agegrp)
label define agelbl 1 "15-24" ///
  2 "25-34" 3 "35-54" 4 "55+"
label values agegrp agelbl

tabulate agegrp
```

EXTRA The table header is long. Add `label variable agegrp "Age group"` and rerun.

YOU SHOULD SEE

Four labelled groups adding up to 4,634: the working-age records.

IF NOT

`agegrp already defined`? Rerun the whole file.

Your result

```
.    recode ilo_age (0/14 = .) (15/24 = 1) (25/34 = 2) (35/54 = 3) (55/max = 4), generate(agegrp)
(5,895 differences between ilo_age and agegrp)

.
.    label define agelbl 1 "15-24" 2 "25-34" 3 "35-54" 4 "55+"

.
.    label values agegrp agelbl

.
.    tabulate agegrp
```

RECODE of ilo_age (Age in completed years (top-coded at 85))	Freq.	Percent	Cum.
15-24	721	15.56	15.56
25-34	789	17.03	32.59
35-54	1,658	35.78	68.36
55+	1,466	31.64	100.00
Total	4,634	100.00	

Read the recode output

```
. tabulate agegrp, missing

  RECODE of |
    ilo_age |
     (Age in |
  completed |
      years |
 (top-coded |
    at 85)) |      Freq.     Percent        Cum.
------------+-----------------------------------
      15-24 |        721       12.23       12.23
      25-34 |        789       13.38       25.61
      35-54 |      1,658       28.13       53.74
        55+ |      1,466       24.87       78.61
          . |      1,261       21.39      100.00
------------+-----------------------------------
      Total |      5,895      100.00
```

1. the automatic label is long: add your own with `label variable`
2. 15–24: 721 persons
3. 0 to 14 became missing, as the rule said

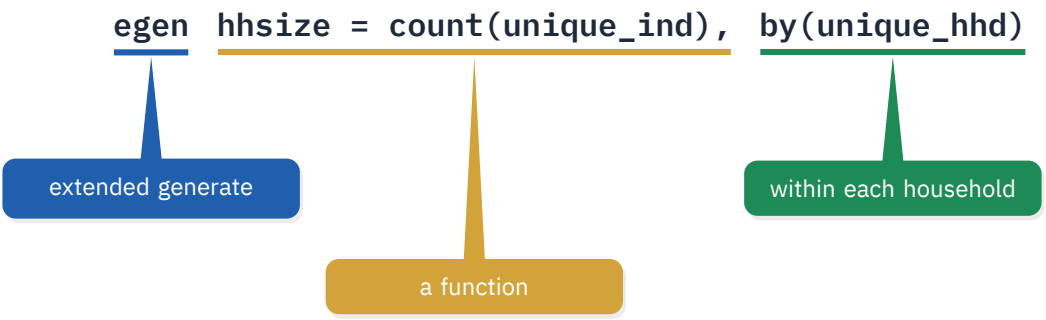
RULE

Recode into a new variable.

Keep the original answers. Next quarter someone will want different groups.

```
…, generate(agegrp)
```

Calculate across rows or within groups:



Household size from the person file

```
egen hhsize = count(unique_ind), by(unique_hhd)
```



Every member of a household gets the same number.

YOUR TURN · TASK 46

Add household size

DO

Add and run:

```
egen hhsize = count(unique_ind), by(unique_hhd)
list unique_hhd ppno ilo_age hhsize in 1/6
```

YOU SHOULD SEE

The first households, each member with the same `hhsize`.

IF NOT

`hhsize already defined`? Rerun the whole file.

EXTRA Count the employed in each household: `egen n_employed = total(employed), by(unique_hhd)`.

Your result

```
. egen hhsize = count(unique_ind), by(unique_hhd

. list unique_hhd ppno ilo_age hhsize in 1/6
```

	uniqu~hd	ppno	ilo_age	hhsize
1.	112110	1	60	3
2.	112110	2	31	3
3.	112110	3	9	3
4.	11211	1	47	3
5.	11211	2	48	3
6.	11211	3	17	3

Household size on every person

```
. list unique_hhd ppno hhsize in 1/6

     +--------------------------+
     | uniqu~hd   ppno   hhsize |
     |--------------------------|
  1. |   352339      2        4 |
  2. |    51411      1        1 |
  3. |    52341      3        3 |
  4. |    35249      3        4 |
  5. |   444020      3        4 |
     |--------------------------|
  6. |   414564      1        5 |
     +--------------------------+
. summarize hhsize

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
      hhsize |      5,895     4.14877    1.925069          1         11
```

1

1. every person carries the size of their household
2. mean 4.1, largest 11

Employed persons per household

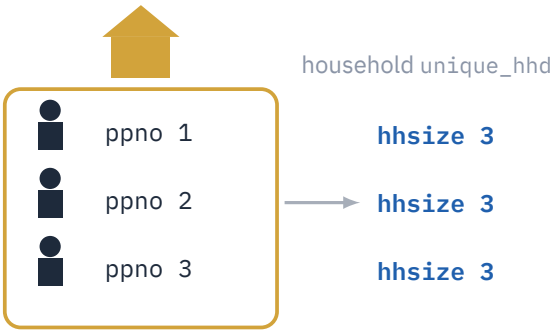
```
. tabulate n_employed

 n_employed |      Freq.     Percent        Cum.
------------+-----------------------------------
          0 |        447        7.58        7.58
          1 |      1,535       26.04       33.62
          2 |      2,133       36.18       69.80
          3 |      1,121       19.02       88.82
          4 |        464        7.87       96.69
          5 |        161        2.73       99.42
          6 |         27        0.46       99.88
          7 |          7        0.12      100.00
------------+-----------------------------------
      Total |      5,895      100.00
```

1

1. 447 persons live in a household where nobody is employed

One household, one value on every row



Across the columns of one row

var1	var2	var3
2	·	4



```
rowtotal(var1 var2 var3)  = 6
rowmean(var1 var2 var3)   = 3
```

Both skip the missing value: the total treats it as 0, the mean leaves it out.

YOUR TURN · TASK 47

Count difficulties per person

DO

1. Check the codes: `label list DIF_SIGHT`
2. Count the answers with a lot of difficulty or cannot do it:

```
egen n_dif = anycount(dif_*), values(3 4)
```

EXTRA `tabulate n_dif`

YOU SHOULD SEE

`n_dif` from 0 to 6: in how many of the six domains each person has serious difficulty.

IF NOT

Different codes in the label list? Use those codes in `values()`.

A week has 168 hours.
Anything above is a code, not a statistic.

YOUR TURN · TASK 48

Clean the hours

DO

1. Look: `tabulate wkt_mj_acthrs if wkt_mj_acthrs > 168 & !missing(wkt_mj_acthrs)`
2. Read the label: `codebook wkt_mj_acthrs`
3. Recode: `mvdecode wkt_mj_acthrs, mv(997=.a)`

EXTRA Do the same for `wkt_mj_ushrs_rep`.

YOU SHOULD SEE

One row above 168: **DON'T KNOW**, 13 records. After the recode, `summarize` shows a maximum of 168.

IF NOT

Other codes above 168? Add them to `mv()` with their own letter.

Your result

```
. tabulate wkt_mj_acthrs if wkt_mj_acthrs > 168 & !missing(wkt_mj_acthrs)
```

Actual hours in main job	Freq.	Percent	Cum.
DON'T KNOW	13	100.00	100.00
Total	13	100.00	

```
. codebook wkt_mj_acthrs
```

wkt_mj_acthrs — **Actual hours in main job**

```
                Type: Numeric (float)
               Label: WKT_MJ_ACTHRS, but 79 nonmissing values are not labeled

               Range: [0,997]                       Units: .1
      Unique values: 80                         Missing .: 2,877/5,895

            Examples: 35
                      45
                      .
                      .
```

```
. mvdecode wkt_mj_acthrs, mv(997=.a)
wkt_mj_act~s: 13 missing values generated

. summarize wkt_mj_acthrs
```

Variable	Obs	Mean	Std. dev.	Min	Max
wkt_mj_act~s	3,005	39.3421	12.62638	0	168

Part 6 on one page

CHEAT CARD · KEEP THIS ONE

`rename old new`	a better name, nothing else changes
`generate v = expression if ...`	a new variable
`replace v = value if ...`	change values of an existing one
`(a == 1)`	true = 1, false = 0
`recode v (a/b = 1) ..., generate(new)`	groups, kept in a new variable
`egen v = count(x), by(g)`	a group count on every row
`egen v = total(x), by(g)`	a group total on every row
`egen v = rowtotal(a b c)`	across variables, missing as 0

Practice on your own

PRACTICE SET · practice_D1.do, PART 6

- CORE — P1 Create `female` (1 woman, 0 man) and tabulate it.
- CORE — P2 Create `youth`: 1 for ages 15–24, 0 for 25 and over, missing under 15.
- CORE — P3 Number of children under 15 in each household, on every row.
- EXTENSION — P4 Recode `ilo_age` into 10-year groups starting at 15; label them.
- CHALLENGE — P5 Share of household members employed, on every row (two egen lines).

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

You have built NBS's core labour indicators from the raw answers.

lost? open **catchup_D1_S6.do**



7

DAY 1 · PART 7

Text and time, the Stata way

Numbers and text look alike

a number

1121

black in the Data Editor

can be added, averaged, compared

a string (text)

1121101

red in the Data Editor

a label, not a quantity

Both examples are from your file: `ea_q` and `unique_ind` (Part 1, Data Editor).

YOUR TURN · TASK 49

Find text variables

DO

Type:

```
describe mig_bth_cnty mig_ctz
tabulate mig_ctz
```

YOU SHOULD SEE

Storage type `str3` and `str2`: text. Citizenship holds answer letters: `a`, `ab`, `b`, `c`.

IF NOT

`variable not found`? Check `cnty` at the end of the first name.

Your result

```
. describe mig_bth_cnty mig_ctz

Variable      Storage   Display    Value
    name         type    format    label      Variable label
------------------------------------------------------------------------------
mig_bth_cnty    str3    %9s                   Name of country of birth
mig_ctz         str2    %9s                   Citizenship status

. tabulate mig_ctz

Citizenship |
     status |      Freq.     Percent        Cum.
------------+-----------------------------------
          a |      5,548       94.11       94.11
         ab |         55        0.93       95.05
          b |        291        4.94       99.98
          c |          1        0.02      100.00
------------+-----------------------------------
      Total |      5,895      100.00
```

Text variables

```
. describe mig_bth_cnty mig_ctz

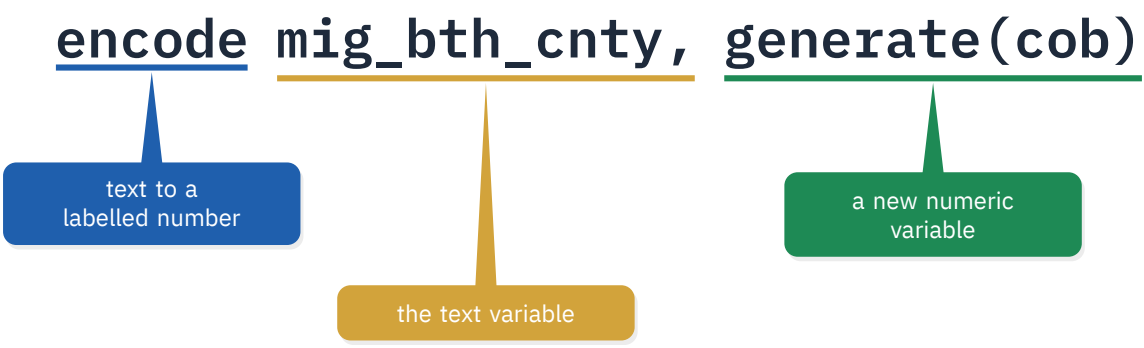
Variable      Storage   Display    Value
    name         type    format    label      Variable label
------------------------------------------------------------------------------
mig_bth_cnty    str3    %9s                   Name of country of birth (1)
mig_ctz         str2    %9s                   Citizenship status (2)

. tabulate mig_ctz

Citizenship |
     status |      Freq.     Percent        Cum.
------------+-----------------------------------
          a |      5,548       94.11       94.11 (3)
         ab |         55        0.93       95.05
          b |        291        4.94       99.98
          c |          1        0.02      100.00
------------+-----------------------------------
      Total |      5,895      100.00
```

1. str3: up to 3 characters, a country code
2. str2: citizenship stored as letters
3. letters cannot be averaged or put in a model: encode them

Turn text codes into categories:



Each text becomes a number; the text becomes its label.

YOUR TURN · TASK 50

Encode country of birth

DO

Add to `my_day1.do` and run:

```
encode mig_bth_cnty, generate(cob)
describe cob
```

YOU SHOULD SEE

`cob`: numeric (`long`), with a value label called `cob`.

IF NOT

`cob already defined`? Rerun the whole file.

EXTRA `tabulate cob`: same countries, now in a proper table.

COMMON ERROR

Encoding a number

```
. encode ilo_sex, generate(sexstr)
not possible with numeric variable
```

WHY

`encode` turns text into labelled numbers. `ilo_sex` is a number already.

FIX

Use `encode` only on `str` variables; check the type with `describe` first.

Work on the text itself

```
generate idlen = strlen(unique_ind)
generate cnty_clean = strtrim(upper(mig_bth_cnty))
```

number of characters

capitals, spaces removed at both ends

`tabulate idlen` shows the IDs of different lengths you met in Part 3.

A Stata date is a number of days



The number is what Stata stores. A **format** shows it as a date.

Build a date from its parts:

```
generate ref_date = mdy(int_month, 15, 2026)
```

a new variable

month, day,
year to a date

Then show it as a date:

```
format ref_date %td
```

how to show it

as a day: 15jan2026

YOUR TURN · TASK 51

Create the reference date

DO

Add and run:

```
generate ref_date = mdy(int_month, 15, 2026)
format ref_date %td
list int_month ref_date in 1/5
```

YOU SHOULD SEE

The month name next to a real date such as `15jan2026`.

IF NOT

A number like `24121`? The `format` line did not run.

EXTRA `generate ref_month = mofd(ref_date)`, then `format ref_month %tm`.

Your result

```
. generate ref_date = mdy(int_month, 15, 2026)

. format ref_date %td

. list int_month ref_date in 1/5

     +-----------------------+
     | int_mo~h    ref_date  |
     |-----------------------|
  1. |  January   15jan2026  |
  2. |  January   15jan2026  |
  3. |  January   15jan2026  |
  4. | February   15feb2026  |
  5. | February   15feb2026  |
     +-----------------------+

. display mdy(1,15,2026)
24121

.
```

The same dates, before and after the format

NUMBER OF DAYS

```
. list int_month ref_date in 1/3

     +----------------------+
     | int_mo~h   ref_date |
     |----------------------|
  1. |    March      24180 |
  2. | February      24152 |
  3. |    March      24180 |
     +----------------------+
```

FORMAT %td

```
. list int_month ref_date in 1/3

     +----------------------+
     | int_mo~h    ref_date |
     |----------------------|
  1. |    March   15mar2026 |
  2. | February   15feb2026 |
  3. |    March   15mar2026 |
     +----------------------+
```

24180 is the number of days since 1 January 1960: 15 March 2026. The format changes the display, never the value.

QUICK CHECK

What does `display mdy(2, 30, 2026)` show?

A. 2 March 2026

B. a missing value: there is no 30 February

C. an error message

ANSWER

What does `display mdy(2, 30, 2026)` show?

A 2 March 2026

B a missing value: there is no 30 February

C an error message

`mdy()` returns missing for a date that does not exist, without a warning. Check new dates with `count if missing(ref_date)`.

RULE

A date that reads 24121 is correct, but unformatted.

Add the format. The value does not change; only what you see.

```
format ref_date %td
```

Text codes are now categories and months are real dates.

lost? open **catchup_D1_S7.do**



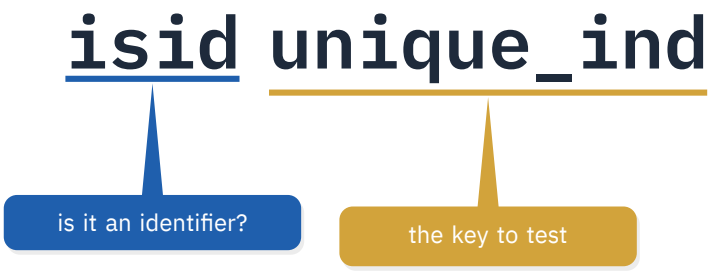
8

DAY 1 · PART 8

One person, one row

Every NBS file has a key: the variable that names each row once.

Check the key:



Silent if every value appears once. Red if not.

YOUR TURN · TASK 52

Test the person key

DO

Type:

```
isid unique_ind
```

and press **Enter**.

YOU SHOULD SEE

Look carefully at what Stata says.

IF NOT

Red `variable not found`? Check the spelling: `unique_ind`.

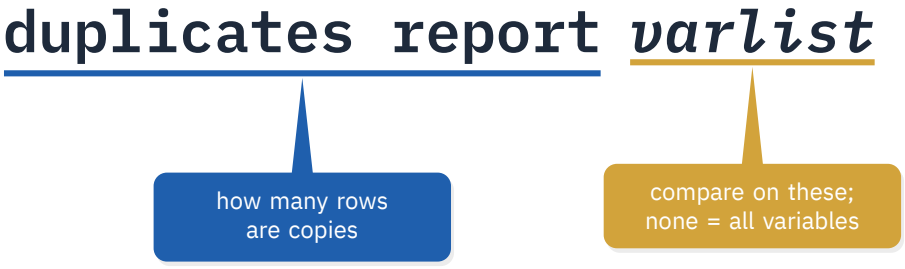
Your result

```
. isid unique_ind
variable unique_ind does not uniquely identify the observations
r(459);

.
```

Good. Stata found a problem before it reached a table.

Count the repeated rows:



YOUR TURN · TASK 53

How many copies?

DO

Type:

```
duplicates report
duplicates report unique_ind
```

YOU SHOULD SEE

Two small tables with **copies**, **observations** and **surplus**.

IF NOT

Red text? Check the spelling: `duplicates`, then `report`.

EXTRA Compare the two tables: are the repeated IDs complete copies?

Your result

```
. duplicates report

Duplicates in terms of all variables
```

Copies	Observations	Surplus
1	**5881**	**0**
2	**14**	**7**

```
. duplicates report unique_ind

Duplicates in terms of unique_ind
```

Copies	Observations	Surplus
1	**5881**	**0**
2	**14**	**7**

Read duplicates report

```
. duplicates report

Duplicates in terms of all variables

--------------------------------------
   Copies | Observations       Surplus
----------+---------------------------
        1 |         5881             0
        2 |           14             7
--------------------------------------
```

1. 5,881 rows appear once
2. 14 rows come in pairs: 7 pairs, so 7 surplus rows

QUICK CHECK

duplicates report shows 14 observations with 2 copies. How many rows does duplicates drop remove?

A 14

B 7

C 2

ANSWER

duplicates report shows 14 observations with 2 copies. How many rows does duplicates drop remove?

A 14

B 7

C 2

14 rows form 7 pairs. One row of each pair stays, so 7 are removed: the Surplus column.

Mark the copies so you can see them:

```
duplicates tag, generate(dup)
```

mark repeated rows

0 = unique, 1 or more = has copies

Then: `list unique_ind ilo_sex ilo_age if dup`

YOUR TURN · TASK 54

Look at the copies

DO

Add and run:

```
duplicates tag, generate(dup)
sort unique_ind
list unique_ind ilo_sex ilo_age if dup
```

YOU SHOULD SEE

Pairs of rows, the same ID and the same answers.

IF NOT

Too many rows? Check `if dup` at the end of the list line.

Your result

```
. duplicates tag, generate(dup)

Duplicates in terms of all variables

. sort unique_ind

. list unique_ind ilo_sex ilo_age if dup
```

	uniqu~nd	ilo_sex	ilo_age
867.	1443513	2 - Female	38
868.	1443513	2 - Female	38
1479.	1814211	2 - Female	63
1480.	1814211	2 - Female	63
2153.	2116383	2 - Female	24
2154.	2116383	2 - Female	24
3470.	2629162	1 - Male	62
3471.	2629162	1 - Male	62
4465.	4140382	1 - Male	54
4466.	4140382	1 - Male	54
4644.	4147272	1 - Male	40
4645.	4147272	1 - Male	40
5172.	4437531	2 - Female	58
5173.	4437531	2 - Female	58

The same person, entered twice

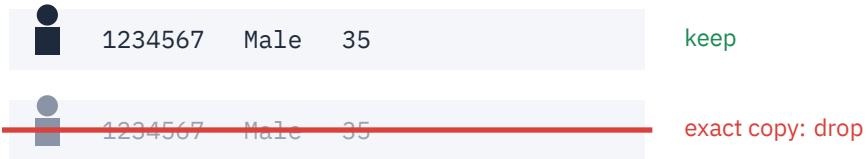


Illustration: the values show the layout, not a real duplicate.

Remove exact copies:

```
duplicates drop
```

keeps one row of each set of exact copies

RULE

Drop exact copies only.

The same ID with different answers is a question for the field team, not for a command.

YOUR TURN · TASK 55

Drop the copies, test again

DO

Add and run:

```
duplicates drop
isid unique_ind
count
```

YOU SHOULD SEE

`(7 observations deleted)`, a silent `isid`, and **5,888**.

IF NOT

`isid` still red? Check that `duplicates drop` ran without a varlist.

Your result

```
. duplicates drop

Duplicates in terms of all variables

(7 observations deleted)

. isid unique_ind

. count
  5,888

. isid unique_hhd ppno

.
```

After the drop

```
. duplicates drop

Duplicates in terms of all variables

(7 observations deleted) 1

. isid unique_ind

. count
  5,888 2
```

1. 7 observations deleted
2. 5,888 persons; `isid` silent

Parts 7 and 8 on one page

CHEAT CARD · KEEP THIS ONE

`encode s, generate(n)`	text to labelled numbers
`mdy(m, d, y)`	a date as a day count
`format v %td`	show a day count as a date
`mofd(d) + format %tm`	the month of a date
`isid key`	silent if the key is unique
`duplicates report [key]`	how many copies
`duplicates tag, generate(dup)`	flag the copies
`duplicates drop`	remove exact copies only

YOUR TURN · TASK 56

Check the household key too

DO

Add and run:

```
isid unique_hhd ppno
```

YOU SHOULD SEE

Nothing: household and person number together name each row once.

IF NOT

Red? Tell the trainer: that would be a finding for NBS.

Your file now has a guaranteed key, checked by Stata.

lost? open **catchup_D1_S8.do**

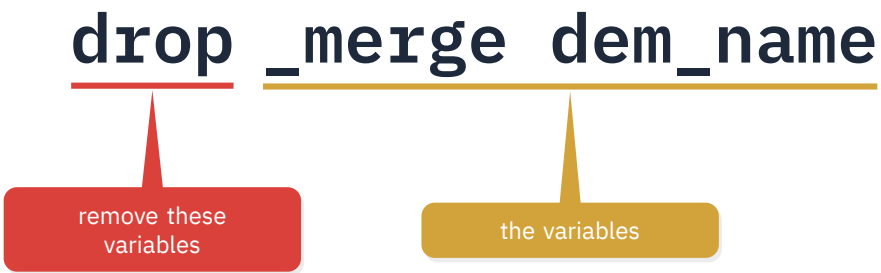


9

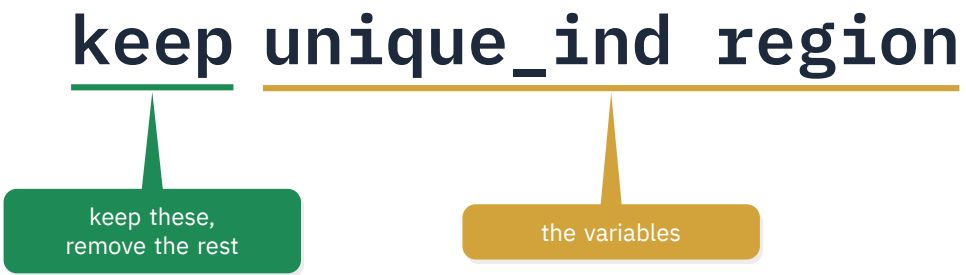
DAY 1 · PART 9

Today’s product

Remove variables:



Or keep only some:



YOUR TURN · TASK 57

Drop four variables, with a reason each

DO

Add to `my_day1.do`:

```
* _merge: left over from a merge
* dem_name: empty
* mjj_occ_tle: free text
* ilo_wgt: 1 for everyone
drop _merge dem_name mjj_occ_tle ///
  ilo_wgt dup
```

YOU SHOULD SEE

Five names gone from the Variables window.

IF NOT

`variable dup not found`? You did not tag duplicates in Part 8: remove `dup` from the line.

A weight of 1 is not a weight

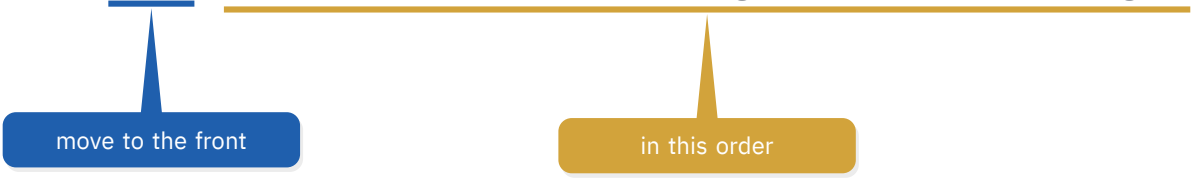
```
drop _merge dem_name mjj_occ_tle ilo_wgt
```



Day 3 explains what `wgt_final` does to every estimate.

Put the key and design variables first:

```
order  unique_ind unique_hhd ppno region district psu ea wgt_*
```



Same data, smaller file:

`compress`

the smallest
storage type
that keeps every value

What compress reports

```
. drop _merge dem_name mjj_occ_tle ilo_wgt dup

. compress
  variable employed was float now byte (1)
  variable unemployed was float now byte
  variable in_lf was float now byte
  variable hhsize was float now byte
  variable n_employed was float now byte
  variable cob was long now byte
  variable ref_date was float now int
  variable ref_month was float now int
  (129,536 bytes saved) (2)
```

1. each variable stored in the smallest type that holds its values
2. 129,536 bytes saved; no value changes

YOUR TURN · TASK 58

Final checks before saving

DO

Add and run:

```
isid unique_ind
assert inlist(region, 1, 2, 3, 4, 5, 6, 7)
```

YOU SHOULD SEE

Nothing: both checks are silent.

IF NOT

Red? Read the first line: which check failed?

YOUR TURN · TASK 59

Save today’s product

DO

Add and run:

```
label data "NBS LFS Q1 2026, teaching file"
compress
save "data/lfs_teach.dta", replace
```

YOU SHOULD SEE

file data/lfs_teach.dta saved.

IF NOT

file already exists? Add , replace.

COMMON ERROR

Stata will not overwrite without being told

```
. save "data/lfs_raw.dta"
file data/lfs_raw.dta already exists
```

WHY

Without `replace`, `save` never overwrites. Here that protects the raw file NBS delivered.

FIX

Save the clean file under a new name, `lfs_teach.dta`. Use `replace` only for your own products.

Your result

```
. compress
  variable employed was float now byte
  variable unemployed was float now byte
  variable hhsize was float now byte
  variable cob was long now byte
  variable ref_date was float now int
  (82,432 bytes saved)

. save "data/lfs_teach.dta", replace
file data/lfs_teach.dta saved
```

The analysis file

```
. describe, short

Contains data from _trainer/capture/teach_test.dta
 Observations:         5,888                  NBS LFS Q1 2026, teaching file
    Variables:            96                  7 Oct 2026 14:41
Sorted by: ilo_sex
```

1. 5,888 persons, with the data label
2. 96 variables: 90 from NBS, minus 4 dropped, plus the ones you made

QUICK CHECK

Why save the clean file as lfs_teach.dta and never over lfs_raw.dta?

A. the raw file is the record of what NBS delivered, and the do-file rebuilds the clean one

B. Stata cannot save over a file

C. the clean file is smaller

ANSWER

Why save the clean file as lfs_teach.dta and never over lfs_raw.dta?

A. the raw file is the record of what NBS delivered, and the do-file rebuilds the clean one

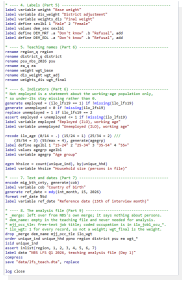
B. Stata cannot save over a file

C. the clean file is smaller

Raw data stay as received. The do-file is the recipe that turns them into the analysis file, so anyone can check or repeat the cleaning.

Your do-file, top to bottom

```
version 17   clear all   cd …  log using …
use "data/lfs_raw.dta", clear
duplicates drop   isid unique_ind
mvdecode …  label …  rename …
generate …  recode …  egen …
encode …  drop …  assert …
save "data/lfs_teach.dta", replace
log close
```



Move the duplicates lines straight after use.

Household size then counts persons, not copies.

YOUR TURN · TASK 60

The real test

DO

1. Save `my_day1.do`.
2. Close Stata completely.
3. Open Stata, open `my_day1.do`, press **Ctrl + D**.

YOU SHOULD SEE

The whole day runs again, with no red, ending with the save and `end of do-file`.

IF NOT

Red somewhere? Read its first line, fix that line, run again from the top.

Your result

```
. save "data/lfs_teach.dta", replace
file data/lfs_teach.dta saved


.
. log close
      name:  <unnamed>
       log:  C:\NBS_Stata\logs/day1.log
  log type:  text
 closed on:   6 Oct 2026, 17:23:18
-------------------------------------------------------------------------------


.
end of do-file
```

That is reproducibility: same files, same do-file, same result.

Today’s product is done: a clean, labelled LFS analysis file, rebuilt by one do-file.

lost? open **catchup_D1_S9.do**

Lab 2 · Clean the job variables

THE SITUATION

The employment section needs clean job variables for the hours and social-protection tables: no codes in hours, readable labels, a 0/1 indicator.

OPEN

Copy `lab_D1_2.do` from the USB `day1` folder into `C:\NBS_Stata\day1`. It starts from `lfs_teach.dta`.

SEVEN TASKS

CORE	Tasks 1 to 5
EXTENSION	Task 6
CHALLENGE	Task 7

CORRECT WHEN

The whole file runs with no red text. The lab does not change `lfs_teach.dta`.

The lab file in your editor

```
* ------------------------------------------------------------------
* TASK 1 (Core)  Settings, diary, data
*   Settings and diary as in every do-file today, then open
*   data/lfs_teach.dta. Log name: logs/lab_D1_2.log
* ------------------------------------------------------------------
* >>> YOUR CODE


* CHECK Task 1 (silent = correct)
assert c(version) == 17
assert c(N) == 5888
isid unique_ind


* ------------------------------------------------------------------
* TASK 2 (Core)  Usual hours without codes
*   wkt_mj_ushrs_rep holds usual weekly hours in the main job.
*   Find the code above 168 (codebook), check its label, and turn it
*   into .a with mvdecode, as you did for actual hours.
* ------------------------------------------------------------------
* >>> YOUR CODE


* CHECK Task 2 (silent = correct)
assert wkt_mj_ushrs_rep <= 168 | missing(wkt_mj_ushrs_rep)
```

Green note when your last check is silent. Then the Challenge task.

Day 1, done

YOU CAN NOW

- work from a do-file with a log
- inspect any NBS file
- turn codes into labelled missing values
- label variables and values
- build indicators and groups
- guarantee the key of a file

TODAY'S PRODUCT

`lfs_teach.dta`

TOMORROW

Merge the three months and the household file, then build the tables pack in Excel and Word.

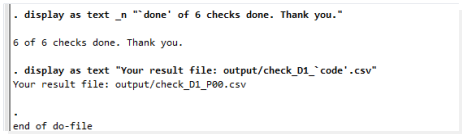
One product a day

- Day 5 · one-button release
- Day 4 · analytical note
- Day 3 · official estimates
- Day 2 · tables pack
- **Day 1 · clean analysis file** — done

Daily check

1. Open `check_D1.do` from the USB `day1` folder.
2. Type your **participant code** on the first line.
3. One sentence: what was unclear today?
4. Save and run with **Ctrl + D**.

Stata prints many lines; read the last one: how many of 6 checks are done. It is not a grade.



```
. display as text _n "`done' of 6 checks done. Thank you."

6 of 6 checks done. Thank you.

. display as text "Your result file: output/check_D1_`code'.csv"
Your result file: output/check_D1_P00.csv

.
end of do-file
```

Thank you. Tomorrow starts with the file you saved today.